Global–Local 3-D Convolutional Transformer Network for Hyperspectral Image Classification

Wenchao Qi, Changping Huang, Yibo Wang, Xia Zhang, Weiwei Sun, *Senior Member, IEEE*, and Lifu Zhang, *Senior Member, IEEE*

***Abstract*— Benefiting from powerful feature extraction capabilities, convolutional neural networks (CNNs) have gained prominence in hyperspectral image (HSI) classification. Nevertheless, with restricted receptive fields of convolution kernels, CNN-based methods fail to learn complex characteristics of long-range sequences. Meanwhile, the vision transformer allows us to learn long-range dependencies in a global view, but local region features are ignored. To overcome these limitations, we propose a novel method entitled global–local 3-D convolutional transformer network (GTCT), where 3-D convolution is embedded in a dual-branch transformer to simultaneously capture global–local associations in both the spectral and spatial domains. In particular, the global–local spectral convolutional transformer (GECT) is designed to exploit global spectral sequence signatures and local spectral relationships between bands. Symmetrically, the global–local spatial convolutional transformer (GACT) is devised to exploit local spatial context features and global interactions among different pixels. In addition, multiscale global–local spectral–spatial information is adaptively fused with trainable weights by the weighted multiscale spectral–spatial feature interaction (WMSFI) module. It is worth noting that a spectral–spatial global attention mechanism (SSGAM) is incorporated into multihead convolutional attention to further integrate discriminative spectral–spatial information. Extensive experiments on four HSI datasets, including the GF-5 and ZY1-02D satellite HSIs, demonstrate the superiority of the proposed GTCT method over other state-of-the-art algorithms with fewer parameters and lower floating-point operations (FLOPs) in practical applications.**

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***Index Terms*— Convolutional token embedding, convolutional vision transformer (ViT), global–local information, hyperspectral image (HSI) classification, multihead spectral–spatial attention.**

I. Introduction

CHARACTERIZED by hundreds of narrow spectral channels ranging from the visible and near-infrared region to the short-wave infrared region, hyperspectral images (HSIs) contain contiguous spectral information and diverse spatial morphology features. HSIs play an important role in many remote sensing applications, such as environmental monitoring [1], precision agriculture [2], and target detection [3]. Being the cornerstone of hyperspectral data analysis, HSI classification, which assigns each pixel to a certain label or category using discriminative features of different materials, is a critical task among various practical applications.

Over the past few years, numerous machine learning approaches based on feature engineering and model training have been developed for HSI classification [4]. Although the two-step classification techniques have been skillfully utilized in HSI classification, these classifiers (e.g., support vector machine [5] and random forest [6]) are considered shallow models with limited and insufficient feature extraction capabilities [7]. Recently, deep learning models have attracted more attention in the field of HSI classification and achieved remarkable success. Compared with traditional classification methods with hand-crafted features, the deep learning methods can hierarchically capture high-level features and improve HSI classification performance end to end. It can be roughly grouped into three main streams: spectral-based methods, spatial-based methods, and spectral–spatial joint methods [4]. Spectral-based methods typically utilize pixel spectrum as the input of well-known backbone networks, such as stacked autoencoders (SAEs) [8], deep belief networks (DBNs) [9], 1-D convolutional neural networks (CNNs) [10], recurrent neural networks (RNNs) [11], and 1-D generative adversarial networks (GANs) [12], to classify HSIs. However, these methods mainly exploit the vector-based spectral signatures, resulting in information loss and insufficient utilization of the spatial texture features of HSIs. In this context, with an image patch as the input, spatial-based methods have been developed to further improve classification performance. In general, PCA is usually applied for dimensionality reduction of hyperspectral data [13]. Then, the image patches, including the central pixel and its adjacent pixels, are input to a 2-D CNN framework. Combined with PCA and 2-D CNN, the spatial-based methods enable to exploit of representative

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features effectively at a low computational cost. However, these methods emphasize the utilization of spatial context information, ignoring the deep spectral feature extraction. Moreover, spectral and spatial features are extracted separately, failing to exploit 3-D spectral–spatial features from HSIs simultaneously and efficiently. Therefore, deep learning methods using 3-D CNNs are naturally developed and employed for spectral–spatial classification of hyperspectral cube data [14], [15]. However, there are still some limitations to CNN-based methods, which are summarized as follows.

1) To avoid overfitting and accuracy decline, CNNs typically require sufficient samples to achieve impressive classification performance. It is difficult to make a good tradeoff between limited training samples and extensive model parameters.
2) CNNs overly focus on the spatial context information and fail to make full use of diagnostic features and subtle discrepancies along the spectral dimension, which has been proven to be crucial and significant for recognizing fine-grained ground objects.
3) CNNs adopt static convolution filters to treat all spectral bands and spatial pixels equally regardless of the redundant information and interferential background classes, barely heightening the valuable representations and sensitivity of features.
4) With the limited receptive field of local and fixed convolution kernels, these CNN-based methods fail to model the middle- and long-range dependencies of numerous spectral bands, thus ignoring the global relationship information contained in patches.

Very recently, vision transformers (ViTs) have almost taken over the field of computer vision (CV) [16] and are preliminarily applied for HSI classification [17]. Originating in the field of natural language processing (NLP), ViTs are more effective at capturing detailed information of long-range spectral bands in parallel through a multihead self-attention (MHSA) mechanism. It not only gets rid of the limitation of fixed convolution kernel but also avoids gradient vanishing of RNNs, such as long short-term memory networks (LSTMs) and gated recurrent unit networks (GRUs) [18]. Similar to the use of words as tokens in the field of NLP, discrete spectral bands or nonoverlapping image patches are usually treated as tokens for HSI classification. Specifically, MHSA highlights the interesting and important sequence features while suppressing insignificant information by matrix multiplication and addition operations. And a position encoding module has been designed to record the coarse spatial information of HSI tokens, whose global relationships are repeatedly extracted by several transformer blocks in the meantime. In addition to pure transformer-based methods, to make use of the strengths of CNN and transformer, several hybrid CNN-transformer approaches have also been investigated for HSI classification [19], [20], [21]. In general, the transformer is primarily responsible for extracting long-range information from spectral sequences, while spatial context features are extracted by CNNs [22]. However, there also exist some limitations, which hinder the further improvement of HSI classification algorithms.

1) Due to the lack of inherent good properties in CNNs, such as local receptive field and distortion invariance, ViTs fail in extracting the local representations and high-level features effectively for HSI classification.
2) The operations of flattened mapping and linear projection greatly distort the spatial geometrical characteristics of various ground objects. Although the position embedding roughly records coarse-grained location information, the diverse spatial context features of HSIs cannot be accurately and inadequately represented.
3) Simply improving transformer-based models designed for processing natural images in the CV field, is not a good fit for HSI classification with lots of narrow spectral bands directly, where MHSA may lead to spectral overdispersion in the global range and insufficient use of local spectral signatures of high-dimensional HSIs.
4) By concatenating separate convolutional modules and transformer encoders directly, existing hybrid CNN-transformer methods focus on introducing inductive bias into the transformer for spectral feature extraction, rarely convolving tokens to thoroughly capture global–local information in both the spatial and spectral domains.

Based on the above-mentioned analysis, from the dual perspectives of spectral–spatial feature extraction and global–local information mining, this work focuses on the effective integration and fusion of spectral–spatial features, global–local features, and multiscale features. We strive to ensure that spatial features are taken into account during spectral feature extraction and global information is considered during local information mining, and vice versa. Specifically, inspired by the benefits of 3-D CNN and transformer, we devise a novel global–local 3-D convolutional transformer network (GTCT), which consists of three modules, namely, the global–local spectral convolutional transformer (GECT), the global–local spatial convolutional transformer (GACT), and the weighted multiscale spectral–spatial feature interaction (WMSFI). To the best of our knowledge, we are the first to introduce 3-D convolution operations to transformers in both spectral and spatial dimensions instead of 2-D depthwise convolution operations and present a dual-branch 3-D convolutional transformer network to learn spectral–spatial representations together in a global–local view. It is worth mentioning that in the overall network framework, each feature extraction branch is embedded with pure 3-D convolution operations without patch flattening and linear projection for HSI classification. The main contributions of this work are summarized as follows.

1) GECT is designed to capture more discriminative spectral features, including long-range dependencies, of the whole spectral sequence and local relationships between several bands along the spectral dimension while remaining intact spatial information.
2) GACT integrates the distortion invariance of 3-D CNN and the advantages of a transformer in modeling global spatial position relationships. It extracts local spatial context features and global interactions among different

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pixels in the HSI cube while avoiding the loss of spectral information.

3) To effectively integrate spectral–spatial features for collaborative classification, the spectral–spatial global attention mechanism (SSGAM) and a WMSFI module are developed to adaptively integrate weighted multiattribute features from different feature extraction backbones and branches at multiple scales.
4) Extensive comparison and ablation experiments on both aerial and satellite HSIs demonstrate the superiority of the proposed GTCT method over other state-of-the-art classification algorithms with fewer parameters and lower FLOPs.

The remainder of this article is organized as follows. Section II reviews the related work. The proposed GTCT network is elaborated on in Section III. Section IV illustrates the HSI datasets, experimental configurations, and results. Section V summarizes the work and provides a brief outlook for future work.

II. Related Work

A. CNN and Attention Mechanism for HSI Classification

Owing to the fine properties of convolution operations including weight sharing and translation equivariance, a series of CNN-based methods have been extensively investigated for HSI classification and achieved markable performance. From 2014 to 2016, researchers mainly applied 1-D CNN and 2-D CNN to extract spectral and spatial features separately. Hu et al. [10] used a five-layer 1-D CNN to capture spectral features of HSI directly, yielding classification improvements over the SVM classifier. After dimensionality reduction by randomized PCA, Makantasis et al. [23] fed the principal components of HSI into a 2-D CNN and made classification using a multilayer perception (MLP) classifier. However, these approaches emphasize only partial use of HSI features, with inadequate extraction of 3-D spectral–spatial features. In this context, many methods using 3-D CNNs have been naturally developed to capture representative features in both spectral and spatial dimensions simultaneously [24], [25], [26], [27], [28]. Without feature preprocessing, Zhong et al. [15] designed a spectral–spatial residual network (SSRN) to learn high-level joint features consecutively through two convolutional branches with residual connections. Similarly, Xu et al. [29] explored the use of a dual-channel residual network (DCRN) to efficiently utilize spectral–spatial information from mislabeled samples and achieved noise-robust classification performance. In addition, in [30], a deformable HSI classification framework (DHCN) was proposed to overcome the deficiencies of the fixed receptive field and capture deformable features from complex spatial contexts of HSI. Then, a densely connected pyramidal dilated convolutional network (PDCN) was proposed in [31] to eliminate blind spots in the receptive field and extract more comprehensive spectral–spatial features for HSI classification. However, existing CNN-based methods ignore the different importance of spatial pixels and unequal contributions of spectral bands, where convolution kernels with limited receptive fields are content independent, yielding inaccurate identification of ground objects with local contextual similarity and large-scale variation.

Thereafter, with the properties of emphasizing important information while suppressing insignificant features, the attention mechanism has been widely combined with CNN architectures [32], [33], [34]. As improved variants of CNN-based methods, the dual-branch spectral and spatial attention networks, such as double-branch multi-attention (DBMA) in [35] and double-branch dual-attention (DBDA) in [36], have been proposed successively to focus on these informative spectral bands and spatial pixels of HSI. Then, an attention-based adaptive spectral–spatial kernel (A^2S^2K) was employed to boost the classification performance of the residual network. In [32], based on the attention mechanism, a hierarchical residual network (HResNetAM) was proposed to adaptively set weights for multiscale spatial and spectral features in parallel. Tang et al. [37] further combined four 3-D octave convolution blocks with the dual-branch spatial–spectral attention module (DOCN) to mine features. Moreover, Zhu et al. [38] introduced a residual spectral–spatial attention network (RSSAN) to find informative pixels and bands, while weakening features with less contribution to HSI classification. Based on the spatial attention mechanism, Zhang et al. [34] integrated a homogeneous pixel detection module (HPDM) into the head of the grouped convolution blocks to efficiently obtain diagnostic features of continuous nonoverlapping subbands. However, these attention-based approaches are essentially improved variants of the CNN-based methods, which suffer from the inherent limitations of local convolution kernels. It also highlights local features and ignores global information, failing to model long-range dependencies of spectral sequences and spatial pixels.

B. HSI Classification Using Vision Transformer

Since the competitive performance in multiple NLP and CV tasks, transformer-based backbones have been introduced to the HSI classification task. Unlike CNNs with static filters, now more recently ViT dynamically captures global sequence dependencies and long-range interactions by MHSA with parallel computing. With a global receptive field, He et al. [39] converted pixels into sequences and adopted bidirectional encoder representations from transformers for HSI classification. Hong et al. [18] and He et al. [40] flattened patches as the input sequence vectors of the transformer to capture global spectral relationships of HSIs. On the contrary, Xue et al. [41] obtained local subpatch sequences by spatial partition restore module and applied a local transformer to capture spatial attention weights among related pixels. With unfold operation and linear projection, Tan et al. [42] introduced a transformer-iN-transformer module to extract both global and local features of EMP cubes for HSI classification. Similarly, Liu et al. [43] also proposed an HSI transformer-iN-transformer method for mapping coastal wetlands, where local features of the superpixel sequences and global features of the mini-patch sequences were obtained by transformer block separately. Combined with contrastive learning, Hu et al. [44] reshaped patches into 1-D sequences and fed the tokens obtained by a fully connected layer to the transformer model

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with position embedding for HSI classification. Gao et al. [45] designed a spatial–spectral ViT to extract sequential relationships of spatial distribution and spectral response from HSI, respectively, where the flattened patches and spectra were linearly mapped as input vectors of the transformer. It can be seen that these previous works mainly adopt transformers to learn the intensive interactions among abundant tokens by multiple self-attention blocks, suffering from slow inference speed and heavy memory footprint. These methods also fail to fully utilize the rich spatial features of HSIs.

C. Vision Transformer Combined With Convolution for HSI Classification

To benefit from the complementary advantages of CNN and transformer, many researchers concentrate on the hybrid CNN-transformer approaches for HSI classification. To this end, CTN has been introduced in [19] to capture spatial position information by center position encoding and obtain local–global features through a convolutional transformer embedded with 2-D convolution operations. Yang et al. [46] explored an HSI transformer (HiT) network and sought for spatial–spectral representations by adaptive 3-D convolutional projection along the spectral dimension and 2-D depthwise convolutions. In [22], with dimensionality reduction using PCA, a spectral–spatial feature tokenization transformer (SSFTT) was proposed, which first combined 2-D and 3-D convolution operations to extract low-level features, followed by the transformer with a Gaussian weighted feature tokenizer module. The work of [21] connected the CNN and transformer directly in both serial and parallel modes to make full use of HSI information. In [47], the spectral–spatial transformer network (SSTN) was investigated, which focused on the interactions between nonlocal features through the spatial attention module and spectral association module. Inspired by CvT in [48] that introduced convolutions to ViTs, Liang et al. [49] grouped the representation of tokens, including channels and pixels, via evenly sampling and embedded 1-D and 2-D convolution operations to multihead contextual self-attention separately. However, most of these methods draw on the experience that is mainly suitable for processing natural images, which are quite different from HSIs with low spatial resolution. How to effectively integrate the excellent locally contextual mining capability of CNN and the global modeling capability of transformer in spectral and spatial dimensions simultaneously, as well as the adaptive fusion of spectral–spatial features with multiple attributes and scales, remains challenging [50].

In this work, in order to make full use of the strengths of CNN and transformer, we investigate a novel GTCT to effectively capture the full spectrum sequence characteristics of ground objects while taking into account multiscale spatial information of HSIs. The proposed GTCT consists of two feature extraction modules and an adaptive fusion module for multiscale spectral–spatial features. Three-dimension spectral–spatial convolution operations are embedded to learn local inductive bias within transformers, effectively integrating the full spectrum sequence features and local spatial features of HSI in different domains.

III. Proposed Method

In this section, the proposed GTCT method for HSI classification is introduced, which includes the details of the GECT module, GACT module, WMSFI module, and SSGAM block.

A. Overview of the Proposed GTCT Network

In this work, we aim to develop a novel GTCT for the refined vegetation classification of HSIs. As delineated in Fig. 1, the proposed GTCT mainly comprises three modules: the GECT module in the spectral branch, the GACT module in the spatial branch, and the WMSFI module in the spectral–spatial branch. The GECT module is first designed to simultaneously capture global sequence information and local convolution characteristics along the spectral dimension, which includes two spectral convolutional (SeConv) token embedding blocks and two spectral convolutional transformer (Spectral ConvFormer) encoders. Subsequently, similar token embedding blocks and encoders adopted in the GACT module can improve the extraction capability of global–local spatial features, where both the long-range dependencies between pixels and local spatial relationships are obtained by transformers and convolution kernels in the spatial dimension. Finally, the global–local spectral and spatial features obtained from the spectral and spatial branches are convolved by different kernels to obtain multiscale features and then adaptively fused by multiplying a set of learnable weights in the process of model training. It is worth noting that, as the key components with convolutional projection and convolutional token embedding, the multihead spectral convolutional attention (MHECA) and multihead spatial convolutional attention (MHACA) with SSGAM are designed and embedded in GECT and GACT modules, respectively. Thereafter, by associating with layer normalization, residual connection as well as feed-forward, the encoders of spectral convolutional transformer and spatial convolutional transformer are proposed accordingly. Table I reports the detailed definitions of the proposed GTCT method.

B. Spectral–Spatial Global Attention Mechanism

As shown in Fig. 2, in order to effectively capture discriminative features across three dimensions of HSI simultaneously, an SSGAM block is designed. Unlike GAM embedded with 2-D convolution operations [51], the SSGAM utilizes different 3-D convolution kernels to generate attention weights from a global perspective, extracting important spectral information while remaining spatial features, and vice versa. Without pooling and linear flattening operations, the SSGAM also avoids information loss and numerous computations in the 3-D space of HSIs. Specifically, for the input feature map $F_{\text{input}} \in \mathbb{R}^{H\times W\times C}$, where (H, W) represents the spatial size and C is the number of channels, the intermediate feature map F_{inter} and output feature map F_{output} can be formulated as

$$F_{\text{inter}} = T_{\text{spe}}(F_{\text{input}}) \otimes F_{\text{input}} \tag{1}$$

$$F_{\text{output}} = T_{\text{spa}}(F_{\text{inter}}) \otimes F_{\text{inter}} \tag{2}$$

where the T_{spe} and T_{spa} are the spectral and spatial feature maps, respectively, and $\otimes$ denotes elementwise multiplication.

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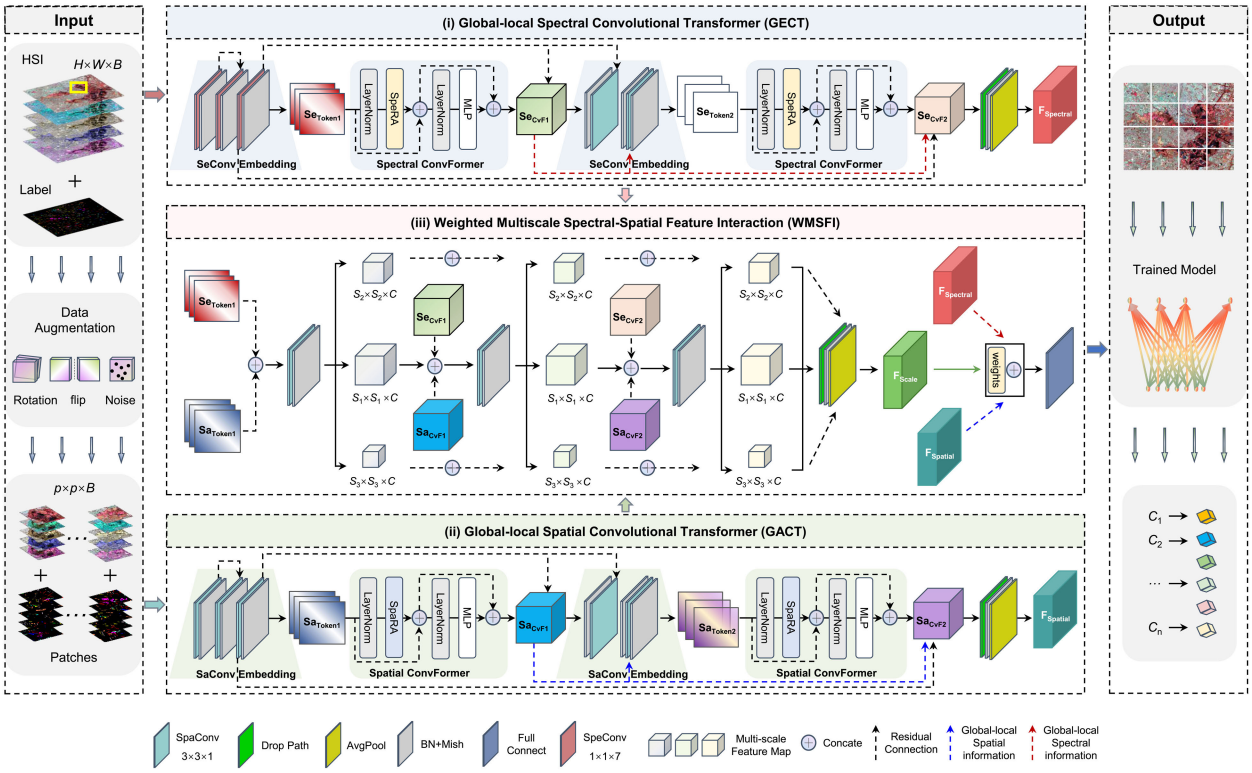


Fig. 1. Overall pipeline of the proposed GTCT method.

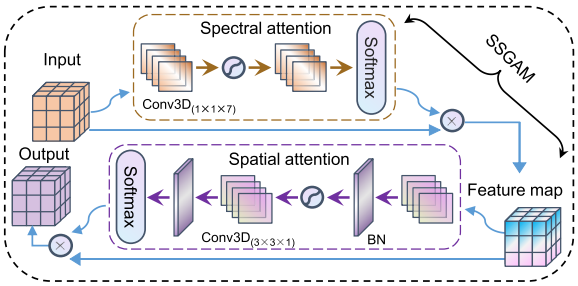


Fig. 2. SSGAM.

In the spectral attention subblock, we first employ two spectral filters of $1 \times 1 \times 7$ size to convolve the input feature map without changing the feature dimension. These spectral features are then fed into the softmax function to generate spectral attention weights, which are multiplied with the original input to refine features. In the spatial attention subblock, two spatial filters of $3 \times 3 \times 1$ size are first applied to extract spatial information from the intermediate feature map. With batch normalization (BN) layers, the outcomes of previous layers are further standardized to accelerate the gradient convergence. Then, combined with the softmax function, relatively significant spatial information is captured. Accordingly, the joint spectral–spatial features are comprehensively obtained by the above-described sequential attention subblocks in SSGAM.

C. Global–Local Spectral Convolutional Transformer

Focusing on the long-range dependencies across all spectral bands, the vanilla ViT fails to extract local spectral features, while convolution kernels with inductive bias pay more attention to the local information contained in these adjacent bands efficiently. Taking advantage of both transformer and convolution, the GECT aims to capture global and local features of HSI across the spectral dimension. As shown in Fig. 1, unlike the original ViT architecture, the GECT module is built with two types of blocks, namely, spectral convolutional token embedding and spectral convolutional transformer. The overall architecture of the global–local convolutional transformer is illustrated in Fig. 3(a). When the operations of convolutional embedding and residual attention in Fig. 3(a) are executed along the spectral dimension, and the GECT module is formed.

1) Spectral Convolutional Token Embedding: The goal of spectral convolutional token embedding is to make full use of local spectral features while maintaining abundant spatial detailed information of HSI across three dimensions. Instead of converting HSI patches to spectral feature sequences, we employ different 3-D convolution kernels on the input patches or features to extract joint spectral–spatial features simultaneously without dimensional compression. By setting different parameters for convolution kernels, spatial or spectral characteristics are emphasized.

Specifically, as shown in Fig. 1, given the original HSI data $I \in \mathbb{R}^{H \times W \times B}$, where H and W refer to the height and width of the HSI with B spectral bands. We first crop the original hyperspectral data to obtain image patches with a size of $X \in \mathbb{R}^{p \times p \times B}$. Then, in the spectral convolutional token embedding block of the GECT module, spectral convolution filters with a kernel size of $1 \times 1 \times k$ and a subsampling stride of $1 \times 1 \times s$ are utilized to transform the image patches to feature maps, where b is $((B - k)/s + 1)$. By this means, the number of bands is reduced from B to b with

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TABLE I
ARCHITECTURAL DETAILS OF THE PROPOSED GTCT NETWORK ON IP DATASET

Module	Block	Definition	Input size	Output size
Input		-	1×9×9×200	-
GECT	SeConv Embedding$_1$	$\begin{bmatrix} 2@1\times1\times7Conv3D \\ 1\times1\times97Conv3D \end{bmatrix}$	1×9×9×200	24×9×9×1
	Spetral ConvFormer$_1$	$SpeRA:\begin{cases} LayerNorm \\ MHECA:\begin{bmatrix} 3@1\times1\times7Conv3D \\ Softmax \end{bmatrix}\times2 \\ SSGAM:\begin{bmatrix} 2@1\times1\times7Conv3D \\ 2@3\times3\times1Conv3D \\ 2@Softmax \end{bmatrix} \\ LayerNorm+MLP \end{cases}$	24×9×9×1	24×9×9×1
	SeConv Embedding$_2$	2@3×3×1Conv3D	48×9×9×1	12×9×9×1
	Spetral ConvFormer$_2$	$SpeRA:\begin{cases} LayerNorm \\ MHECA:\begin{bmatrix} 3@1\times1\times7Conv3D \\ Softmax \end{bmatrix}\times2 \\ SSGAM:\begin{bmatrix} 2@1\times1\times7Conv3D \\ 2@3\times3\times1Conv3D \\ 2@Softmax \end{bmatrix} \\ LayerNorm+MLP \end{cases}$	12×9×9×1	12×9×9×1
GACT	SaConv Embedding$_1$	$\begin{bmatrix} 3\times3\times200Conv3D \\ 2@3\times3\times1Conv3D \end{bmatrix}$	1×9×9×200	24×9×9×1
	Spatial ConvFormer$_1$	$SpaRA:\begin{cases} LayerNorm \\ MHACA:\begin{bmatrix} 3@1\times1\times7Conv3D \\ Softmax \end{bmatrix}\times2 \\ SSGAM:\begin{bmatrix} 2@1\times1\times7Conv3D \\ 2@3\times3\times1Conv3D \\ 2@Softmax \end{bmatrix} \\ LayerNorm+MLP \end{cases}$	24×9×9×1	24×9×9×1
	SaConv Embedding2	2@3×3×1Conv3D	48×9×9×1	12×9×9×1
	Spatial ConvFormer$_2$	$SpaRA:\begin{cases} LayerNorm \\ MHACA:\begin{bmatrix} 3@1\times1\times7Conv3D \\ Softmax \end{bmatrix}\times2 \\ SSGAM:\begin{bmatrix} 2@1\times1\times7Conv3D \\ 2@3\times3\times1Conv3D \\ 2@Softmax \end{bmatrix} \\ LayerNorm+MLP \end{cases}$	12×9×9×1	12×9×9×1
WMSFI		$\begin{bmatrix} 3\times3\times1Conv3D \\ 3\times3\times1Conv3D \\ 5\times5\times1Conv3D \end{bmatrix}\times3$	$\begin{bmatrix} 48\times9\times9\times1 \\ 60\times9\times9\times1 \\ 36\times9\times9\times1 \end{bmatrix}$	$\begin{bmatrix} 12\times9\times9\times1 \\ 12\times7\times7\times1 \\ 12\times5\times5\times1 \end{bmatrix}$
Fully Connected Layer		Linear	276×1×1	16×1×1
Softmax				

the computational load decreased. Thereafter, followed by two convolutional layers with a kernel size of $1 \times 1 \times k$ and $1 \times 1 \times b$, spectral convolution tokens can be obtained. In 3-D convolutional layers, the value at position (x, y, z) on the jth feature map in the ith layer can be expressed as

$$v_{i,j}^{xyz} = g\left(b_{i,j} + \sum_{m}\sum_{h=0}^{H_i-1}\sum_{p=0}^{P_i-1}\sum_{t=0}^{T_i-1}\omega_{i,j,m}^{h,p,t} v_{i-1,m}^{(x+h)(y+p)(z+t)}\right) \quad (3)$$

where $g(\cdot)$ represents the activation function, and the height and width of the 3-D convolution kernel in the spatial dimension are denoted by H_i and P_i, respectively. T_i is the kernel size along the spectral dimension. The number of kernels of the ith layer is j, m is the feature cube connected to the current feature cube in the $(i-1)$th layer, the (h, p, t)th value of the kernel connected to the mth feature cube in the preceding layer is denoted $\omega_{i,j,m}^{h,p,t}$, and $b_{i,j}$ is the bias on the jth feature cube in the ith layer. Through three 3-D convolutional layers along the spectral dimension and one residual connection layer in the spectral convolutional token embedding block, spectral convolution tokens T_{spe} can be formulated as follows:

$$\begin{aligned} F_{\text{spe}}^1 &= \text{Conv3}D_{1\times1\times k_1}(X) \\ F_{\text{spe}}^2 &= \text{Conv3}D_{1\times1\times k_2}(F_{\text{spe}}^1) \\ T_{\text{spe}} &= \text{Conv3}D_{1\times1\times k_3}\left(\text{Concat}\left(F_{\text{spe}}^1, F_{\text{spe}}^2\right)\right) \end{aligned} \quad (4)$$

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where k_1, k_2, and k_3 represent the channel sizes of different spectral convolution kernels, respectively. It is worth noting that due to the dimensionality reduction of features, we also use spatial convolution kernels to highlight spatial context information in the second spectral convolutional token embedding block while extracting spectral signatures.

2) Spectral Convolutional Transformer: The spectral convolutional transformer mainly includes spectral residual attention (SpeRA), multilayer perceptron (MLP), and layer normalization. Most importantly, to improve the original MHSA in ViT according to the 3-D spectral–spatial attributes of HSI, the SpeRA is designed, which consists of MHECA, BN, dimensional transition layer, and SSGAM block. The residual structure is also employed within SpeRA to enhance information interactions in the internal layers between different attention blocks. Fig. 3(b) displays the detailed structure of a SpeRA block. As the key component of SpeRA, the MHECA first maps the embedding spectral convolutional tokens into groups of vectors by 3-D convolution operations with spectral kernels of $1 \times 1 \times k$, rather than linear projection, where each group consists of a query (Q), a key (K), and a value (V). Suppose T_{spe} is the input embedding token, Q, K, and V tensors are calculated and reshaped as follows:

$$P_{\text{spe}}^{Q/K/V} = \text{Reshape}\left(\Im_{1\times1\times k}\left(T_{\text{spe}}, W_{\text{spe}}^{Q/K/V}\right) \in \mathbb{R}^{n\times p\times p\times c}\right) \tag{5}$$

where $\Im_{1\times1\times k}(\cdot)$ indicates the 3-D spectral convolution with a kernel size of $1 \times 1 \times k$, and $W_{\text{spe}}^{Q/K/V}$ represents the learnable weight matrix for *Q/K/V* tensors. Also, n, p, and c are channel size, height or width, and spectral dimension of projection vectors, respectively.

After extracting local spectral information by 3-D convolutional layers, three projection vectors are rearranged to produce the following spectral attention map $\text{Attn}_{\text{spe}}^{i}$:

$$P_{\text{spe}}^{Q_i} \in \mathbb{R}^{h\times(p\times p)\times c\times d}, \quad P_{\text{spe}}^{K_i} \in \mathbb{R}^{h\times c\times(p\times p)\times d}$$
$$\text{Attn}_{\text{spe}}^{i} = \text{softmax}\left(\frac{P_{\text{spe}}^{Q_i} P_{\text{spe}}^{K_i\,T}}{\sqrt{d}}\right) \in \mathbb{R}^{h\times c\times c\times d} \tag{6}$$

where $\text{softmax}(\cdot)$ represents the softmax function along the last dimension of input, h and $d = n/h$ are the number of heads and feature dimension in each attention head, respectively. Intuitively, the spectral attention map demonstrates the local spectral association among input tensors, highlighting the importance of different HSI spectral bands. Based on the spectral attention map, the original values are weighed by multiplying matrix V and the attention map in each attention head, which models different and subtle spectral information in an independent feature subspace

$$\text{head}_{\text{spe}}^{i} = P_{\text{spe}}^{V_i}\text{Attn}_{\text{spe}}^{i}, \quad P_{\text{spe}}^{V_i} \in \mathbb{R}^{h\times(p\times p)\times c\times d}. \tag{7}$$

Then, multiple heads are concatenated and calculated in parallel to learn the global spectral features from different subspaces in the MHECA block. The output of the MHECA block can be expressed as

$$\text{MHECA}\left(Q_{\text{spe}}, K_{\text{spe}}, V_{\text{spe}}\right) = \text{Concat}\left(\text{head}_{\text{spe}}^{i}, \text{head}_{\text{spe}}^{2}, \ldots, \text{head}_{\text{spe}}^{h}\right). \tag{8}$$

In summary, depending on the 3-D convolution in token embedding and convolutional projection, the spectral convolutional transformer encoder not only focuses on the spectral correlations between local adjacent bands but also accurately depicts long-range sequence characteristics of all spectral bands. It is noteworthy that the size of the spectral convolution kernel is set to $1 \times 1 \times 7$ in this work.

D. Global–Local Spatial Convolutional Transformer

The vanilla ViT mainly focuses on modeling long-range sequence dependencies in the spectral dimension and fails to extract rich spatial features usually obtained by CNNs. Therefore, the GACT module is designed here, where two convolution-based operations, including convolutional token embedding and convolutional projection, are also introduced to capture global–local spatial features. In contrast to the GECT module, we concentrate on more informative spatial features through a similar structure in the GACT module, which consists of two main blocks, namely, spatial convolutional token embedding and spatial convolutional transformer. It is worth mentioning that we not only focus on extracting spatial features between adjacent pixels within the restricted receptive fields of convolution kernels but also use transformers to extract geometric position information related to all pixels of the entire image patch in a global view.

1) Spatial Convolutional Token Embedding: The spatial convolutional token embedding block aims to capture the geometric position relationships between different pixels by 3-D convolutional layers in the spatial dimension. Based on the global–local 3-D convolutional transformer in Fig. 3(a), we perform the convolutional embedding operations and residual attention calculation in the spatial dimension. Formally, given an image patch $X \in \mathbb{R}^{p\times p\times B}$, a function that convolves the image patch into a spatial token with a kernel size of $l \times l \times 1$ is learned, where $l \times l$ denotes the spatial size of 3-D convolutional kernel. Similar to the spectral convolutional token embedding block, three 3-D convolutional layers along the spatial dimension and one residual connection layer in the spatial convolutional token embedding block are designed, the spatial convolutional tokens T_{spa} can be expressed as follows:

$$\begin{aligned} F_{\text{spa}}^{1} &= \text{Conv3}D_{l_1\times l_1\times 1}(X) \\ F_{\text{spa}}^{2} &= \text{Conv3}D_{l_2\times l_2\times 1}\left(F_{\text{spa}}^{1}\right) \\ T_{\text{spa}} &= \text{Conv3}D_{l_3\times l_3\times 1}\left(\text{Concat}\left(F_{\text{spa}}^{1}, F_{\text{spa}}^{2}\right)\right) \end{aligned} \tag{9}$$

where l_1, l_2, and l_3 stand for the spatial sizes of different convolution kernels, respectively. The spatial convolutional token embedding allows us to extract local spatial contexts without compromising spectral information.

2) Spatial Convolutional Transformer: With similar processing in spectral convolutional transformer encoder, the specific components of the spatial convolutional transformer encoder are illustrated in Fig. 1. In addition to MLP and layer normalization layers in ViT, the spatial residual attention (SpaRA) is developed to enhance the extraction capability of spatial features. In particular, MHACA, BN, dimensional transition layer, and SSGAM block together constitute the

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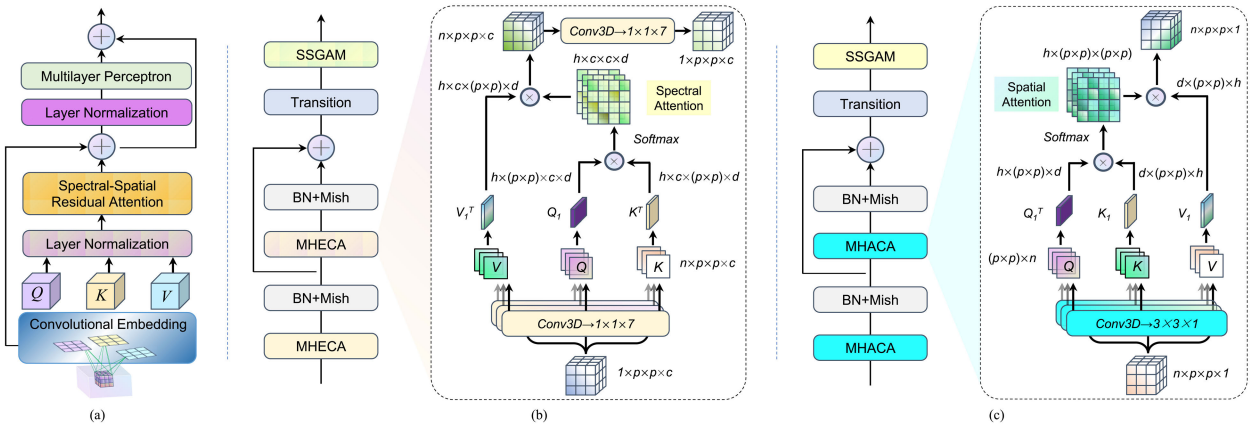


Fig. 3. Global–local 3-D convolutional transformer based on SpeRA and SpaRA. (a) Overall pipeline. (b) SpeRA based on MHECA. (c) SpaRA based on MHACA.

SpaRA block. A skip connection is adopted to smooth model training and regularize the SpaRA block. Fig. 3(c) demonstrates the detailed structure of a spatial residual attention (SpaRA) block. Inspired by the MHSA in ViT, the MHACA block calculates multihead convolutional attention in parallel by 3-D convolution kernels in the spatial domain, aggregating local spatial relationships within tokens and global interactions among all embeddings. In particular, given the spatial embedding token, three 3-D convolutional layers with a kernel size of $l \times l \times 1$ are first employed to map the spatial convolutional tokens into Q, K, and V vectors

$$P_{\text{spa}}^{Q/K/V} = \text{Reshape}\left(\mathscr{G}_{l\times l\times 1}\left(T_{\text{spa}}, W_{\text{spa}}^{Q/K/V}\right) \in \mathbb{R}^{n\times p\times p\times 1}\right) \tag{10}$$

where $\mathscr{G}_{l\times l\times 1}(\cdot)$ denotes the 3-D convolution with a kernel size of $l \times l \times 1$ and a padding operation to keep the spatial dimension unchanged. $W_{\text{spa}}^{Q/K/V}$ represents a learnable weight matrix for three projection tensors. Also, n and p are channel size and height or width, respectively.

After capturing local spatial features by 3-D convolutional layers, the dot-product operation is implemented between Q and K to generate a nonlocal spatial attention map $\text{Attn}_{\text{spa}}^{i}$ among all pixels of the embedding token

$$P_{\text{spa}}^{Q_i} \in \mathbb{R}^{d\times(p\times p)\times h}, \quad P_{\text{spa}}^{K_i} \in \mathbb{R}^{d\times(p\times p)\times h}$$
$$\text{Attn}_{\text{spa}}^{i} = \text{softmax}\left(\frac{P_{\text{spa}}^{Q_i\,T} P_{\text{spa}}^{K_i}}{\sqrt{d_k}}\right) \in \mathbb{R}^{h\times(p\times p)\times(p\times p)} \tag{11}$$

where softmax$(\cdot)$ represents the softmax function along the last dimension of the input. h and $d = n/h$ are the number of heads and feature dimensions in each attention head, respectively. Thereafter, the resulting attention matrix is utilized to weigh how much each pixel contributes to the output representation of each other pixel. And the output of each head is given by the dot-product of projection value tensors and spatial attention matrix. Finally, multiple attention heads make up the MHACA block

$$\text{head}_{\text{spa}}^{i} = P_{\text{spa}}^{V_i}\text{Attn}_{\text{spa}}^{i}, \quad P_{\text{spa}}^{V_i} \in \mathbb{R}^{d\times(p\times p)\times h} \tag{12}$$

$$\text{MHACA}(Q_{\text{spa}}, K_{\text{spa}}, V_{\text{spa}}) = \text{Concat}\left(\text{head}_{\text{spa}}^{i}, \text{head}_{\text{spa}}^{2}, \ldots, \text{head}_{\text{spa}}^{h}\right). \tag{13}$$

It is worth mentioning that the size of the spectral convolution kernel is set to $3 \times 3 \times 1$ in this work. Through these spatial convolutional layers, the expressive capabilities of CNN to extract local spatial information are introduced into the transformer. Thus, the spatial convolutional transformer encoder can not only effectively model the spatial interactions of adjacent pixels but also learn the long-range dependencies among all pixels.

E. Weighted Multiscale Spectral–Spatial Feature Interaction

Due to the influence of different branches and backbones (e.g., CNN and Transformer) for feature extraction, feature distributions are varied. In addition, different ground objects have different spatial scales, and the same ground object also has a variety of spatial sizes. As a result, the WMSFI module is developed for adaptively integrating features with various distributions and multiple scales. As illustrated in the WMSFI module of Fig. 1, these features of various distributions include two aspects: global–local CNN-transformer-based features and local convolutional features. In particular, these global–local spectral and spatial features include $\mathscr{F}_{\text{SpeCvF}}$ and $\mathscr{F}_{\text{SpaCvF}}$ from convolutional transformer encoders, $\mathscr{F}_{\text{GECT}}$ and $\mathscr{F}_{\text{GACT}}$ from the GECT and GACT modules, respectively. However, $\mathscr{F}_{\text{SeCE}}$ and $\mathscr{F}_{\text{SaCE}}$ from convolutional token embedding blocks in both spectral and spatial domains belong to local features. It is worth mentioning that in the process of feature fusion, global features captured by transformer encoders and local features extracted by convolutional token embedding blocks also interact with each other at multiple levels through residual connection layers. Then, based on these interactive global–local features, the features at three different scales are fully exploited through different convolutional layers with changeable kernels in the spatial dimension. Finally, to better integrate heterogeneous spectral–spatial information of ground objects, a weighting vector is designed and dynamically assigned to multiple features during model training

$$\begin{aligned}\mathscr{F}_{\text{Weight}} = \text{Concat}\big(&\eta_1 \cdot \mathscr{F}_{\text{GECT}} + \eta_2 \cdot \mathscr{F}_{\text{GACT}} \\ &+ \eta_3 \cdot \mathscr{F}_{\text{Scale}}^{\text{CE}}(f_{\text{spe}}, f_{\text{spa}}) \\ &+ \eta_4 \cdot \mathscr{F}_{\text{Scale}}^{\text{CvF}_1}(f_{\text{spe}}, f_{\text{spa}}) \\ &+ \eta_5 \cdot \mathscr{F}_{\text{Scale}}^{\text{CvF}_2}(f_{\text{spe}}, f_{\text{spa}})\big)\end{aligned} \tag{14}$$

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TABLE II
GROUND OBJECT CATEGORIES AND NUMBERS OF SAMPLES ON THE FOUR HYPERSPECTRAL DATASETS

Indian Pines (IP)			Xinjiang Cotton (XC)			Anxin Wetland (AW)			Gaofeng Forest (GF)		
No.	Class	Train./Total.	No.	Class	Train./Total.	No.	Class	Train./Total.	No.	Class	Train./Total.
1	Alfalfa	3/46	1	Cotton	309/7733	1	Corn	197/6595	1	China-fir	251/8371
2	CornN	71/1428	2	Grape	306/7668	2	CornN	159/5316	2	Masson pine	188/6272
3	CornM	41/830	3	Woods	113/2840	3	Woods	93/3104	3	Slash Pine	129/4316
4	Corn	11/237	4	Buildings	298/7462	4	Vegetables	8/282	4	E.grandis x	232/7757
5	GrassP	24/483	5	Corn	92/2324	5	Water	104/3494	5	E.urophylla	214/7161
6	GrassT	36/730	6	Greenhouse	13/326	6	Rice	178/5964	6	C. hystrix	251/8372
7	GrassM	3/28	7	Grassland	133/3328	7	Buildings	153/5112	7	Oil tea	122/4091
8	HayW	23/478	8	Water	105/2646	8	Reed	115/3864	8	Lao mytilaria	249/8324
9	Oats	3/20	9	Bare Soil	194/4872	9	Sorghum	16/545	9	Other broadleaf	242/8090
10	SoybeanN	48/972	10	Rice	35/881	10	Bare Soil	93/3130	10	Road	179/5998
11	SoybeanM	122/2455	11	Nursery	220/5518	11	Greenhouse	28/948	11	Cutting blank	219/7324
12	SoybeanC	29/593	12	Peach	37/938	12	Lotus	117/3913	12	Buildings	3/68
13	Wheat	10/205	13	Flat peach	74/1871	13	Grassland	51/1710			
14	Woods	63/1265	14	Road	46/1169						
15	Buildings	19/386	15	Wetland	109/2741						
16	Stone	4/93	16	Alfalfa	11/297						
	Total	510/10249		Total	2095/52614		Total	1312/43977		Total	2279/76144

where η and $\mathscr{F}_{\text{Weight}}$ denote different weights and weighted features, $\mathscr{F}_{\text{GECT}}$ and $\mathscr{F}_{\text{GACT}}$ are the dual-branch spectral–spatial features, $\mathscr{F}^{\text{CE}}_{\text{Scale}}(\cdot)$ represents multiscale features from convolutional embedding block. $\mathscr{F}^{\text{CvF}_1}_{\text{Scale}}(\cdot)$ and $\mathscr{F}^{\text{CvF}_2}_{\text{Scale}}(\cdot)$ stand for multiscale features from two convolutional transformer encoders. The f_{spe} and f_{spa} refer to the features from spectral and spatial branches, respectively. It is noteworthy that features with three spatial scales S_1, S_2, and S_3 are generated in this work.

To train the network, the Polyloss is introduced instead of cross-entropy loss, which can be regarded as a linear combination of polynomial functions. Different from the most common cross-entropy loss, the Polyloss makes it easy to adjust the importance of different polynomial bases according to the target task and dataset, leading to maximal gain while requiring minimal hyperparameter tuning with much more flexible forms. From the perspective of Taylor expansion, the Polyloss function can be formulated as follows:

$$\begin{aligned}\mathscr{L}_{\text{Poly}}(Y,P) &= \mathscr{L}_{\text{CE}} + \varepsilon(1-p_{i,c}) \\ &= -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(p_{i,c}) + \varepsilon(1-p_{i,c}) \quad (15)\end{aligned}$$

where $\mathscr{L}_{\text{CE}}$ is the cross-entropy loss, Y and P denote the ground truth and predicted value of each pixel, respectively. The $y_{i,c}$ and ε refer to the c-th element of the label Y and perturbation factor respectively, where $p_{i,c}$ is the probability of the pixel belonging to c-th category. The C and N represent the number of categories and training samples, respectively. Through the softmax classify function, the probability distribution is predicted, where the input sample belongs to a certain category with the largest probability value.

IV. EXPERIMENTS

In this section, four hyperspectral datasets, including one widely used benchmark hyperspectral dataset of Indian Pines (IP), one airborne hyperspectral dataset of the Gaofeng Forest (GF), and two satellite hyperspectral datasets of the Xinjiang Cotton (XC) and Anxin Wetland (AW) are leveraged to evaluate the robustness and generalization capability of the proposed GTCT method in practical applications for fine classification of various vegetation. The specific ground object types and the number of training and total samples are illustrated in Table II. The number of validation samples is identical to the number of training samples, and the rest samples are testing samples.

A. Data Description

1) Indian Pines: Gathered by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor [52] over an agricultural site in Northwest Indiana in 1992, it has a spatial resolution of 20 m per pixel, covering 145 × 145 pixels. After removing bad bands and water absorption bands, the remaining 200 spectral bands at intervals of 10 nm ranging from 400 to 2500 nm were used for HSI classification. A total of 16 types of ground objects are labeled in this dataset.

2) Gaofeng Forest: It was obtained by the AISA Eagle II diffraction grating push-broom HSIr carried by the airborne LiCHy system [53]. This dataset was collected on January 13 and January 30, 2018, at a flight altitude of approximately 1000 m, which depicts a forest vegetation area with many tree species. It has a spatial resolution of 1 m with 1340 × 853 pixels and a spectral resolution of 3.3 nm with 125 bands ranging from 400 to 1000 nm, respectively. The 12 ground-cover categories are analyzed in classification experiments.

3) Anxin Wetland: It was captured by the visible-shortwave infrared advanced HSIr (AHSI) onboard Gaofen-5 (GF-5) satellite over Anxin County, Hebei Province, China, in 2018. And the GF-5 satellite is an important scientific research satellite in the Chinese Key Project of the High-Resolution

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Earth Observation System [54]. Characterized by spectral resolutions of 5 nm in visible near-infrared (VNIR) and 10 nm in short-wave infrared (SWIR), this dataset has 255 spectral bands ranging from 400 to 2500 nm after eliminating water absorption bands and noisy bands. As for a large-scale wetland covering an area of 909 km^2, it has a spatial resolution of 30 m per pixel and comprises 965 $\times$ 1047 pixels, where 13 ground-cover categories are mapped.

4) Xinjiang Cotton: It was collected by the latest AHSI carried by ZiYuan1-2D (ZY1-02D) satellite over Shihezi City, Xinjiang Province, China, in 2021, where the ZY1-02D satellite is China's first hyperspectral satellite for civilian service [43]. With a spatial resolution of 30-m and 60-km swath width, it acquires 166 spectral bands ranging from 400 to 2500 nm. It has a spectral resolution of 10 nm in VNIR (400∼1000 nm) and 10 nm in SWIR (1000∼2500 nm), respectively. After data processing, this dataset consists of 122 bands and 2450 $\times$ 1000 pixels. In this large-scene satellite HSI covering 2205 km^2, pixels are labeled into 16 ground-cover categories, among which cotton is the dominant cash crop.

B. Experimental Setup

1) Evaluation Metrices: To evaluate classification performance quantitatively, three widely used metrics, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) are utilized. In addition, from the perspective of qualitative analysis, the classification maps and t-distributed stochastic neighbor embedding (t-SNE) feature visualization results of different methods are displayed.

2) Comparison with Competitive Models: In our performance comparison experiments, three types of recently advanced algorithms are adopted to highlight the superiority of the proposed GTCT method. In particular, four CNN-based methods are considered, including DHCN [30], SSRN [15], DCRN [29], and PDCN [31]. In terms of attention-based CNN methods, models including DBMA [35], DBDA [36], A^2S^2K [55], RSSAN [38], and DOCN [37] are compared. In addition, five transformer-based methods including SPRLT [41], SpeFm [18], SSFTT [22], DMuCA [49], and CvT [48], are used as comparison methods. Unless otherwise specified, the default parameters and structures suggested in the literature are used to make a fair comparison. The details of these methods are as follows.

1) DHCN [30] adopts deformable convolutional layers to overcome the limitations of fixed kernels, adaptively adjusting the shape and size of the kernel according to complex spatial information of HSI. It consists of regular and strided convolutional layers and deformable convolutional sampling layers.
2) SSRN [15] designs spectral and spatial residual blocks to learn discriminative spectral signatures and abundant spatial contexts consecutively, which includes 3-D convolutional layers, residual connection blocks, BN, and rectified linear unit (ReLU) activation functions.
3) For DCRN [29], a DCRN for HSI classification with noisy labels, it contains spectral and spatial feature extraction branches, a feature fusion module, and a noise-robust loss function, where 3-D convolutional layers, BN, RELU activation function, average pooling, and a fully connected layer are used.
4) PDCN [31] consists of a total of three densely pyramidal dilated convolutional (DPDC) blocks and two transition layers. With the RELU activation function, each block comprises a pyramidal dilated convolutional layer, a BN layer, and dense connections.
5) DBMA [35], as a dual-branch multiattention mechanism network, applies channelwise attention and spatialwise attention in two parallel branches to refine the feature map separately. It contains several 3-D convolutional layers with residual connections, max-pooling, and average-pooling operations.
6) For DBDA [36], motivated by DBMA, it used shape transformation and 3-D convolutional instead of max-pooling and average-pooling to output spectral and spatial attention maps. Besides, a BN layer, a dropout layer, and a Mish activation function are also applied in this method.
7) A^2S^2K [55] learns selective 3-D convolution kernels adaptively with adjusted receptive field, capturing discriminative spectral–spatial information with an efficient feature recalibration mechanism. In addition to 3-D convolutional layers, the operations, including average-pooling, elementwise addition, and multiplication, are also used in this model.
8) RSSAN [38] combines residual structure and attention mechanism for spectral band selection and the extraction of significant spatial features, suppressing useless information of HSI. This method uses max-pooling, average-pooling, and 2-D convolutional layers to design three attention modules from the perspectives of spatial, spectral, and spectral–spatial dimensions, respectively.
9) DOCN [37] adopts 3-D octave convolution, rather than regular convolution to extract spatial–spectral features, where three modules, including 3D octave convolution, spectral–spatial attention, and information complement, are designed.
10) SpeFm [18] and as a highly flexible backbone framework without any convolution, it integrates groupwise spectral embedding and cross-layer adaptive fusion into the vanilla transformer, which captures spectrally local sequence information from HSI with the pixelwise and patchwise input.
11) SPRLT [41] introduces a local transformer to capture spatial attention weights, with which a spatial partition restore module is designed to model locally detailed spatial discrepancies. Layer normalization, residual connection, and linear mapping are applied individually in the local transformer block.
12) SSFTT [22] designs a Gaussian weighted feature tokenizer to transform features to deep semantic information, where 2-D and 3-D convolutional layers are utilized and independently combined with a transformer for mining shallow spectral–spatial features.
13) DMuCA [49] decouples spatial and spectral contextual attention into two subblocks, which are weighted and

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TABLE III
OA, AA, κ, AND INDIVIDUAL CLASS ACCURACY (%) OF DIFFERENT MODELS ON THE IP DATASET

NO.	CNNs				Attention-based CNNs					Transformer Architectures					
Class	DHCN (2018)	SSRN (2018)	PDCN (2021)	DCRN (2022)	DBMA (2019)	DBDA (2020)	A^2S^2K (2021)	RSSAN (2021)	DOCN (2021)	CvT (2021)	DMuCA (2022)	SpeFm (2022)	SSFTT (2022)	SPRLT (2022)	GTCT (Ours)
C1	89.75	91.11	94.37	92.33	90.62	91.37	90.72	92.31	94.53	96.42	90.37	61.64	84.88	96.21	**100.0**
C2	92.37	96.49	94.97	97.48	91.35	96.34	96.15	86.42	92.78	78.74	86.88	80.68	91.80	92.45	**98.94**
C3	90.02	96.18	96.36	**97.77**	91.96	97.65	95.78	80.24	93.11	81.17	89.38	77.40	93.18	95.29	96.35
C4	90.18	94.91	95.77	96.38	90.51	96.53	93.57	81.20	89.22	86.19	89.07	81.21	88.40	93.11	**99.47**
C5	92.72	99.43	98.68	98.47	96.76	98.43	98.79	97.39	97.47	96.69	99.21	92.95	99.39	99.01	**99.90**
C6	98.34	99.27	99.31	99.07	97.66	99.51	99.04	98.19	98.79	97.96	98.61	92.22	98.04	98.61	**99.62**
C7	76.51	78.65	80.31	70.26	61.88	67.02	79.74	66.14	71.07	70.32	75.40	47.73	85.65	82.75	**100.0**
C8	99.75	99.80	100.0	100.0	99.09	100.0	99.77	98.38	99.23	98.76	99.30	95.06	99.43	99.82	**100.0**
C9	82.22	82.00	88.43	59.40	45.51	68.48	74.76	61.14	62.45	76.99	73.77	48.46	**84.64**	84.42	74.44
C10	81.81	94.61	91.49	94.68	87.25	93.37	93.07	79.40	83.80	81.17	83.91	76.94	87.80	93.56	**96.40**
C11	91.07	98.21	94.49	98.03	94.81	**98.55**	97.78	89.39	93.87	87.53	93.51	86.85	95.99	95.99	98.25
C12	84.30	**97.82**	95.68	97.37	88.26	96.20	93.68	71.72	89.15	76.29	84.22	67.28	89.38	93.75	97.66
C13	97.94	98.20	98.04	98.95	96.25	99.05	97.89	96.61	97.91	97.52	98.40	91.48	98.85	**99.89**	99.68
C14	97.11	98.48	97.16	98.79	97.39	98.94	98.25	96.12	97.58	95.57	97.02	94.04	98.08	98.29	**99.88**
C15	88.04	95.16	92.30	96.23	85.10	96.30	96.32	80.33	86.40	81.64	84.09	79.49	91.31	93.82	**98.70**
C16	91.45	92.49	94.12	93.22	90.08	89.46	92.55	90.46	87.70	87.12	93.26	89.34	89.75	**94.57**	87.08
OA	90.78 (2.08)	97.22 (0.62)	95.35 (0.78)	97.38 (0.55)	92.65 (1.09)	97.13 (0.60)	96.53 (0.76)	87.29 (2.76)	92.77 (2.34)	86.63 (1.70)	91.18 (1.12)	83.92 (0.83)	94.06 (0.50)	95.61 (0.45)	**98.35 (0.14)**
AA	90.22 (3.16)	94.55 (1.53)	94.47 (1.85)	93.03 (1.52)	87.78 (1.16)	92.95 (1.23)	93.62 (1.63)	85.34 (3.52)	89.69 (2.66)	86.88 (1.83)	89.78 (1.56)	78.92 (1.83)	92.28 (1.33)	94.47 (1.34)	**96.65 (0.44)**
κ	89.52 (2.35)	96.83 (0.71)	94.70 (0.90)	97.01 (0.63)	91.64 (1.24)	96.73 (0.68)	96.05 (0.86)	85.57 (3.12)	91.76 (2.66)	84.80 (1.92)	89.97 (1.28)	81.71 (0.93)	93.24 (0.57)	95.00 (0.51)	**98.12 (0.16)**

summed for 3-D attention perception. With operations of 1-D and 2-D depthwise convolution, pointwise and group convolution, the MHSA is generated by concatenating the keys and queries.

14) CvT [48] introduces convolutions into transformers, which are achieved by designing a convolutional token embedding and a convolutional transformer block using a convolutional projection. It applies the operations of 2-D depthwise convolution, 2-D pointwise convolution, and BN in multiple stages.
15) For the proposed GTCT method, incorporated with 3-D convolutional embedding and projection, two different convolutional transformer encoders are specifically designed to focus on spectral–spatial feature extraction. Notably, benefiting from the synthesized advantages of combining CNN and transformer in each feature extraction branch, both global dependencies and local relationships among numerous spectral bands or diverse spatial pixels are effectively exploited. Thereafter, various features with multiple scales are integrated and fused by a weighted spectral–spatial feature interaction module.

3) Implementation Details: Extensive experiments were implemented on the deep learning platform of PyTorch with an Intel Core i9-10900K CPU (at 3.70 GHz), 64-GB RAM, and an NVIDIA GeForce RTX 2070 GPU (8 GB of ROM) server. For all methods, the AdamW optimizer [54] with the dynamic learning rate adjustment using the cosine annealing method is adopted. The batch size and initial learning rate are set to 128 and 5e-4, respectively. The early stopping strategy with loss no longer decreasing for 20 epochs is introduced to save training time and avoid overfitting [36]. Note that random flipping, rotation, and noise addition are adopted until the number of samples per class is increased to the maximum number of samples in a single category. In addition, to obtain more convincing experimental results, ten trials for each algorithm with randomly selected training samples are conducted.

C. Comparison With State of the Arts

Using a variety of deep learning algorithms, we experimentally demonstrate the effectiveness and generalizability of the proposed GTCT method on two aerial hyperspectral datasets (i.e., IP and GF datasets), and two satellite HSIs (i.e., AW and XC datasets). The details of quantitative analysis and intuitive evaluation are as follows.

1) Quantitative Analysis: For quantitative analysis, the average value and standard deviation of OA, AA, and kappa coefficient are summarized in Tables III–VI, where per-class accuracy is also included. The maximum classification accuracy for each category is highlighted in bold in the corresponding row. Through the objective analysis of the experimental results, we draw the following conclusions.

First, algorithmically speaking, our proposed GTCT outperforms other state-of-the-art methods, yielding the highest classification accuracy in OA, AA, and kappa coefficient, respectively. From the perspective of comparison experiments on different hyperspectral datasets, no matter whether using small-scale aerial images (e.g., 8 km^2 of the IP dataset and 1 km^2 of the GF dataset) or large-scene satellite images (e.g., 909 km^2 of the AW dataset and 2205 km^2 of the XC dataset), our proposed GTCT achieves impressive classification performance for refined vegetation classification. And most land-cover categories in HSIs can be effectively identified by the proposed GTCT, which proves the generalizability and robustness of our algorithm.

In particular, the proposed GTCT exhibits excellent performance, especially in recognizing ground objects with

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TABLE IV
OA, AA, κ, AND INDIVIDUAL CLASS ACCURACY (%) OF DIFFERENT MODELS ON THE GF DATASET

NO.	CNNs				Attention-based CNNs					Transformer Architectures					
Class	DHCN (2018)	SSRN (2018)	PDCN (2021)	DCRN (2022)	DBMA (2019)	DBDA (2020)	A^2S^2K (2021)	RSSAN (2021)	DOCN (2021)	CvT (2021)	DMuCA (2022)	SpeFm (2022)	SSFTT (2022)	SPRLT (2022)	GTCT (Ours)
C1	95.90	97.37	95.69	96.91	96.97	95.93	97.63	93.55	95.93	92.62	93.29	95.39	94.91	96.87	**98.18**
C2	97.15	98.95	98.28	98.89	97.96	99.01	99.14	97.15	97.32	97.74	97.29	96.41	97.26	98.87	**99.28**
C3	94.12	97.36	96.77	97.74	97.29	97.71	97.39	92.69	96.30	93.38	93.85	92.80	94.36	97.33	**98.85**
C4	98.09	99.77	99.42	**99.81**	99.74	99.69	99.50	98.39	98.88	85.62	99.06	98.32	97.02	99.45	99.50
C5	94.43	96.40	95.84	96.73	96.62	96.75	97.61	93.48	96.53	91.67	94.00	94.43	96.10	97.08	**98.48**
C6	92.71	97.02	96.36	96.00	94.75	94.72	95.74	91.06	94.46	91.07	93.80	91.15	93.89	95.04	**97.91**
C7	95.52	98.13	97.72	98.88	97.88	98.36	97.66	94.06	97.20	94.35	94.71	96.81	94.75	97.95	**99.84**
C8	99.45	99.64	99.83	**99.95**	99.51	99.79	99.91	99.01	99.51	98.44	99.35	99.35	99.18	99.47	99.83
C9	94.87	96.90	95.16	96.64	95.62	96.48	97.37	92.77	94.72	89.08	94.42	94.06	94.98	96.08	**97.90**
C10	99.77	99.66	99.76	99.62	99.74	**99.86**	99.67	99.69	99.79	99.54	99.79	99.72	99.76	99.85	99.69
C11	99.39	99.52	99.77	99.09	99.52	99.65	99.74	98.96	99.04	97.81	99.10	97.81	99.03	99.66	**99.84**
C12	89.51	93.70	90.91	93.96	94.37	90.68	**96.96**	66.78	73.06	46.31	66.90	57.75	87.96	95.49	96.93
OA	96.43 (0.36)	98.22 (0.13)	97.59 (0.37)	98.12 (0.50)	97.71 (0.34)	97.87 (0.32)	98.30 (0.29)	95.49 (0.51)	97.15 (0.29)	93.28 (0.55)	96.21 (0.30)	95.88 (0.26)	96.51 (0.33)	97.90 (0.21)	**98.96 (0.07)**
AA	95.91 (1.73)	97.87 (0.93)	97.13 (1.20)	97.85 (0.71)	97.50 (0.74)	97.39 (1.13)	98.19 (0.65)	93.13 (1.66)	95.23 (1.56)	89.80 (1.09)	93.80 (2.26)	92.83 (2.22)	95.77 (0.66)	97.76 (0.43)	**98.85 (0.52)**
κ	96.06 (0.40)	98.04 (0.14)	97.34 (0.41)	97.92 (0.55)	97.47 (0.37)	97.64 (0.35)	98.12 (0.32)	95.02 (0.56)	96.85 (0.32)	92.58 (0.61)	95.82 (0.33)	95.45 (0.29)	96.14 (0.36)	97.68 (0.24)	**98.85 (0.08)**

TABLE V
OA, AA, κ, AND INDIVIDUAL CLASS ACCURACY (%) OF DIFFERENT MODELS ON THE AW DATASET

NO.	CNNs				Attention-based CNNs					Transformer Architectures					
Class	DHCN (2018)	SSRN (2018)	PDCN (2021)	DCRN (2022)	DBMA (2019)	DBDA (2020)	A^2S^2K (2021)	RSSAN (2021)	DOCN (2021)	CvT (2021)	DMuCA (2022)	SpeFm (2022)	SSFTT (2022)	SPRLT (2022)	GTCT (Ours)
C1	97.18	97.13	96.35	96.77	96.39	96.41	97.75	96.01	96.22	95.66	96.82	95.59	97.57	97.78	**97.98**
C2	94.96	97.34	96.63	97.40	96.97	96.47	97.32	96.20	95.51	95.25	97.37	96.10	96.75	97.29	**98.25**
C3	95.42	96.22	96.12	95.73	93.78	96.13	**97.20**	92.94	94.28	93.38	94.32	92.85	96.70	95.41	96.50
C4	80.40	89.35	89.49	92.31	83.84	85.89	88.47	73.96	82.93	77.34	81.03	74.00	85.31	88.34	**92.79**
C5	98.78	98.94	98.45	98.72	97.81	98.05	98.21	98.53	98.71	96.94	98.52	98.99	98.65	98.78	**99.20**
C6	97.83	98.72	98.36	99.31	98.22	98.88	98.87	97.39	97.43	97.26	98.04	96.14	98.59	98.54	**99.78**
C7	99.43	99.58	99.51	99.50	98.80	99.32	99.53	98.53	98.79	98.92	99.13	98.55	99.24	99.41	**99.81**
C8	95.28	97.12	96.48	98.62	97.41	97.71	97.18	93.55	98.79	91.38	94.99	89.89	96.87	97.56	**98.24**
C9	95.46	98.57	96.72	**100.0**	98.22	99.62	97.70	94.84	94.60	90.83	94.68	87.14	98.03	97.29	99.85
C10	98.47	98.72	98.37	98.52	97.68	97.89	98.40	97.79	98.45	97.27	98.16	98.56	98.72	**98.96**	98.80
C11	93.52	93.73	96.17	97.72	96.21	96.65	94.54	91.87	94.29	90.58	95.12	86.63	95.43	97.18	**99.12**
C12	99.30	99.55	99.39	99.62	99.01	99.45	99.62	98.41	98.62	99.01	98.68	98.71	99.47	99.20	**99.82**
C13	97.44	97.43	98.08	**98.88**	97.69	98.76	97.34	96.47	96.73	95.07	97.20	94.50	97.53	97.99	96.47
OA	97.11 (0.58)	97.94 (0.25)	97.59 (0.37)	98.24 (0.11)	97.30 (0.30)	97.74 (0.36)	98.05 (0.19)	96.36 (0.60)	96.74 (0.30)	95.80 (0.49)	97.19 (0.23)	95.57 (0.40)	97.88 (0.25)	98.03 (0.19)	**98.65 (0.08)**
AA	95.65 (1.32)	97.11 (0.75)	96.93 (0.79)	97.93 (0.57)	96.31 (0.74)	97.02 (0.74)	97.09 (0.50)	94.35 (0.80)	95.49 (0.49)	93.76 (1.00)	95.70 (0.76)	92.90 (0.59)	96.84 (0.58)	97.21 (0.41)	**98.20 (0.40)**
κ	96.78 (0.65)	97.71 (0.27)	97.31 (0.41)	98.04 (0.12)	96.98 (0.34)	97.48 (0.41)	97.82 (0.21)	95.94 (0.66)	96.36 (0.33)	95.31 (0.55)	96.87 (0.26)	95.06 (0.40)	97.63 (0.28)	97.80 (0.21)	**98.49 (0.09)**

fewer samples, e.g., alfalfa, corn, grass-pasture-mowed, and hay-windrowed on the IP dataset, and vegetables, sorghum, and greenhouse on the AW dataset. It is worth noting that on the challenging IP dataset with limited and imbalanced samples, GTCT shows great advantages in classifying eight of sixteen categories with OA over 99%, among which three categories can be fully identified. Meanwhile, compared with other methods in terms of OA, our GTCT also achieves an improvement with more than 3% on the AW dataset, 5% on the GF dataset, and 7% on the XC dataset, respectively, which provides an effective solution for fine-grained identification of tree species and county-level vegetation mapping.

Furthermore, with features of multiple properties extracted, the classification performance of dual-branch networks is superior to that of sequential models with a single branch, which validates the effectiveness of HSI joint classification using spectral–spatial information. Moreover, the deeper residual networks can distinguish different land cover types more effectively than shallow models. The main reason is that the more hidden layers have stronger modeling capabilities for spatial contextual information and subtle spectral signatures. These statements can almost be concluded from the resulting analyses of both CNN-based and attention-based CNN methods. For instance, DCRN and SSRN, as well as DBDA and A^2S^2K all achieve superior performance than that of the same kind on the four datasets.

Finally, as for the latest transformer-based methods, due to the limited capability to capture local discrepancies, the performance is relatively poorer than that of CNNs when global sequence dependencies are modeled exclusively using the

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TABLE VI

OA, AA, κ, AND INDIVIDUAL CLASS ACCURACY (%) OF DIFFERENT MODELS ON THE XC DATASET

NO.	CNNs				Attention-based CNNs					Transformer Architectures					
Class	DHCN (2018)	SSRN (2018)	PDCN (2021)	DCRN (2022)	DBMA (2019)	DBDA (2020)	A^2S^2K (2021)	RSSAN (2021)	DOCN (2021)	CvT (2021)	DMuCA (2022)	SpeFm (2022)	SSFTT (2022)	SPRLT (2022)	GTCT (Ours)
C1	99.36	**99.66**	99.41	98.61	98.63	99.18	99.63	99.38	99.42	98.84	99.43	99.34	99.57	99.42	99.53
C2	97.00	98.91	98.15	99.34	98.15	99.39	99.06	96.90	96.62	95.55	97.20	93.23	97.87	97.74	**99.48**
C3	89.56	95.09	91.87	95.93	90.46	94.85	94.73	85.01	90.87	83.06	86.51	81.31	92.13	91.63	**96.46**
C4	95.56	98.23	96.49	97.40	97.69	98.34	98.70	95.71	96.78	95.42	96.64	95.08	97.09	98.21	**99.22**
C5	92.48	96.68	91.44	**98.28**	93.75	95.26	96.48	90.36	91.46	85.97	93.03	85.64	94.95	93.91	95.92
C6	81.94	98.90	89.22	99.91	92.77	**99.74**	98.67	71.57	82.39	59.72	76.81	60.40	89.56	92.84	98.98
C7	94.76	98.21	96.51	97.63	96.79	**98.92**	97.88	93.35	93.24	87.02	94.29	89.28	95.37	97.08	98.14
C8	97.44	99.81	97.67	99.54	99.05	98.47	99.22	97.66	98.14	96.67	98.01	98.50	98.18	98.31	**99.91**
C9	95.55	99.17	97.67	**99.54**	98.20	99.18	98.96	95.48	97.27	93.02	97.05	95.03	98.05	98.97	99.43
C10	92.90	99.33	96.25	**99.80**	97.54	99.36	98.97	94.88	95.19	85.98	96.14	86.12	95.25	98.31	99.09
C11	93.89	98.23	97.05	98.76	96.25	98.08	97.62	94.25	94.25	91.83	94.77	90.43	96.72	97.46	**98.97**
C12	74.07	92.74	80.26	98.14	86.67	**96.41**	90.29	64.07	72.35	53.43	69.73	48.20	86.35	82.79	92.78
C13	87.71	96.47	86.80	**99.13**	92.85	96.29	95.24	81.55	84.74	80.25	83.78	74.74	90.00	91.20	98.45
C14	93.43	97.37	92.61	93.62	92.50	92.03	**97.42**	88.87	91.28	85.11	88.55	88.55	91.53	94.30	97.04
C15	99.95	99.91	99.85	99.64	99.79	100.0	99.90	99.85	99.99	98.97	99.95	99.73	99.87	99.94	**100.0**
C16	91.75	97.58	95.28	89.61	88.60	93.54	96.38	93.39	88.01	82.79	92.33	84.54	93.81	95.87	**99.46**
OA	94.86 (0.81)	98.37 (0.28)	96.17 (0.48)	98.38 (0.08)	96.72 (0.35)	98.19 (0.13)	98.19 (0.35)	94.06 (0.88)	95.11 (0.79)	91.56 (0.63)	94.96 (0.55)	91.38 (0.68)	96.62 (0.67)	97.10 (0.29)	**98.80 (0.09)**
AA	92.33 (1.04)	97.89 (0.38)	94.16 (1.14)	97.81 (0.21)	94.98 (0.69)	97.44 (0.39)	97.45 (0.53)	90.14 (1.29)	92.00 (0.85)	85.85 (1.15)	91.51 (0.91)	85.63 (1.26)	94.77 (1.21)	95.50 (0.45)	**98.30 (0.22)**
κ	94.30 (0.91)	98.19 (0.31)	95.75 (0.53)	98.20 (0.09)	96.36 (0.39)	97.99 (0.14)	97.99 (0.39)	93.41 (0.97)	94.58 (0.88)	90.65 (0.70)	94.41 (0.60)	90.44 (0.76)	96.25 (0.74)	96.78 (0.32)	**98.67 (0.10)**

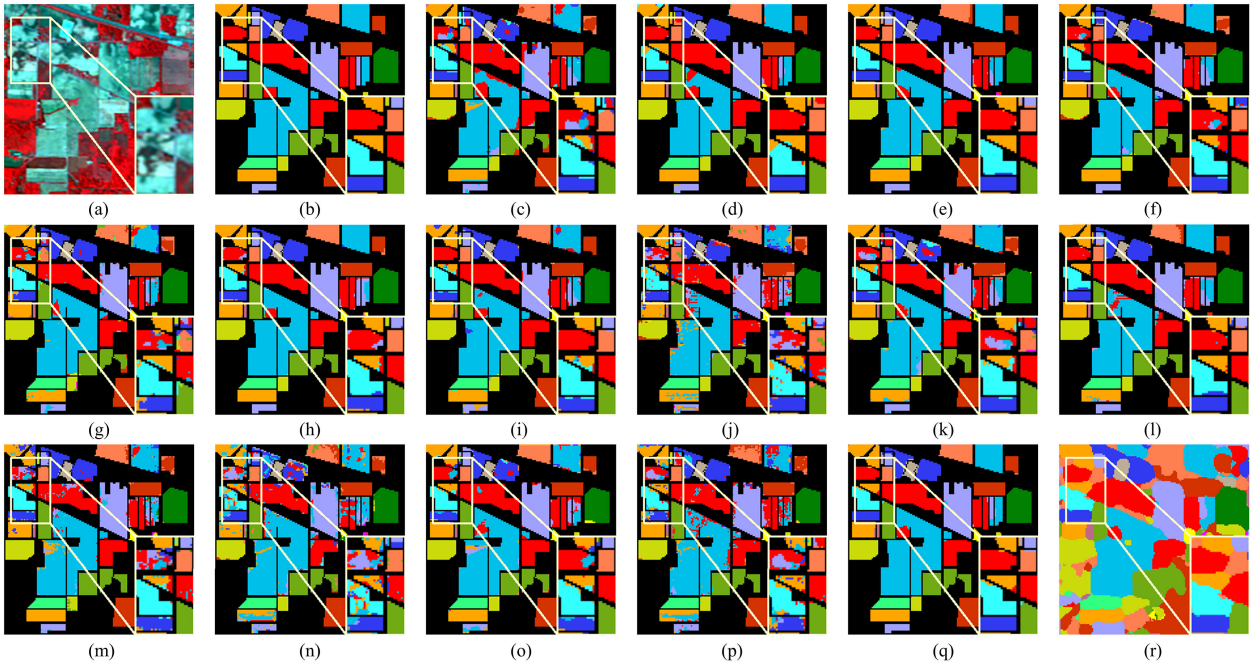


Fig. 4. Classification maps of the IP dataset. (a) False-color image. (b) Ground-truth map. (c) DHCN (OA = 90.78%). (d) SSRN (OA = 97.22%). (e) DCRN (OA = 97.38%). (f) PDCN (OA = 95.35%). (g) DBMA (OA = 92.65%). (h) DBDA (OA = 97.13%). (i) A^2S^2K (OA = 96.53%). (j) RSSAN (OA = 87.29%). (k) DOCN (OA = 92.77%). (l) SPRLT (OA = 95.61%). (m) DMuCA (OA = 91.18%). (n) SpeFm (OA = 83.92%). (o) SSFTT (OA = 94.06%). (p) CvT (OA = 86.63%). (q) Proposed GTCT (OA = 98.35%). (r) All pixel classification map by our proposed GTCT.

self-attention mechanism (i.e., SpeFm). When features are flattened as transformer inputs, it inevitably destroys the abundant spatial information of HSI. However, simply combining convolution and transformer independently can significantly improve classification performance (i.e., SSFTT). Our proposed GTCT leverages a 3-D convolution instead of the 2-D convolution or linear transformation within token embedding and feature projection stages (e.g., SPRLT, CvT, and DMuCA), capturing global and local information simultaneously. Meanwhile, two 3-D convolutional transformer encoders are applied in both spatial and spectral dimensions, where the experimental results of GTCT are better than those of other models that focus only on using spatial information or spectral signatures. In this way, with more representative and discriminative features extracted, our proposed GTCT can achieve the best classification performance compared with other state-of-the-art approaches.

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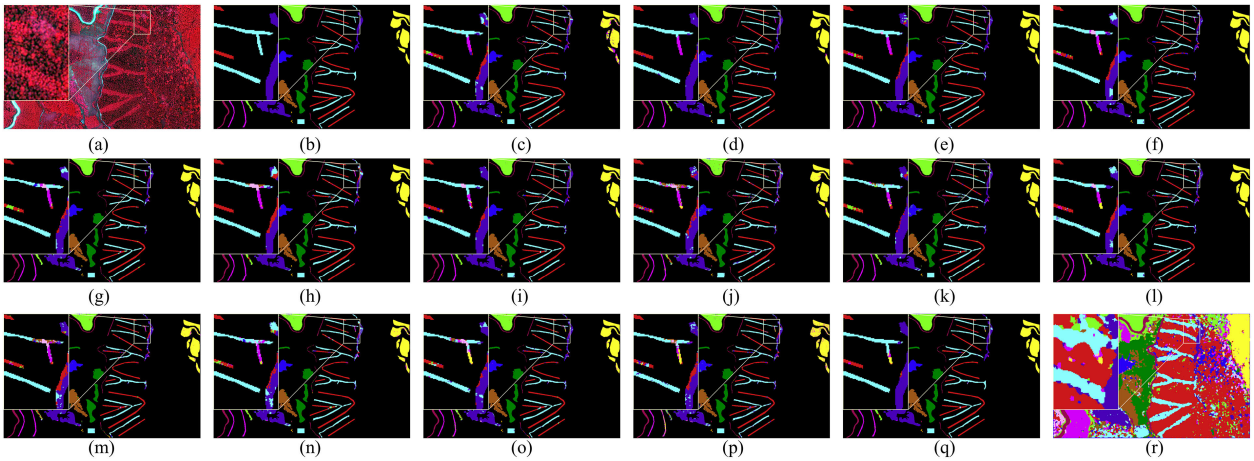


Fig. 5. Classification maps of the GF dataset. (a) False-color image. (b) Ground-truth map. (c) DHCN (OA = 96.43%). (d) SSRN (OA = 98.22%). (e) DCRN (OA = 98.12%). (f) PDCN (OA = 97.59%). (g) DBMA (OA = 97.71%). (h) DBDA (OA = 97.87%). (i) A^2S^2K (OA = 98.30%). (j) RSSAN (OA = 95.49%). (k) DOCN (OA = 97.15%). (l) SPRLT (OA = 97.90%). (m) DMuCA (OA = 96.21%). (n) SpeFm (OA = 95.88%). (o) SSFTT (OA = 96.51%). (p) CvT (OA = 93.28%). (q) Proposed GTCT (OA = 98.96%). (r) All pixel classification map by our proposed GTCT.

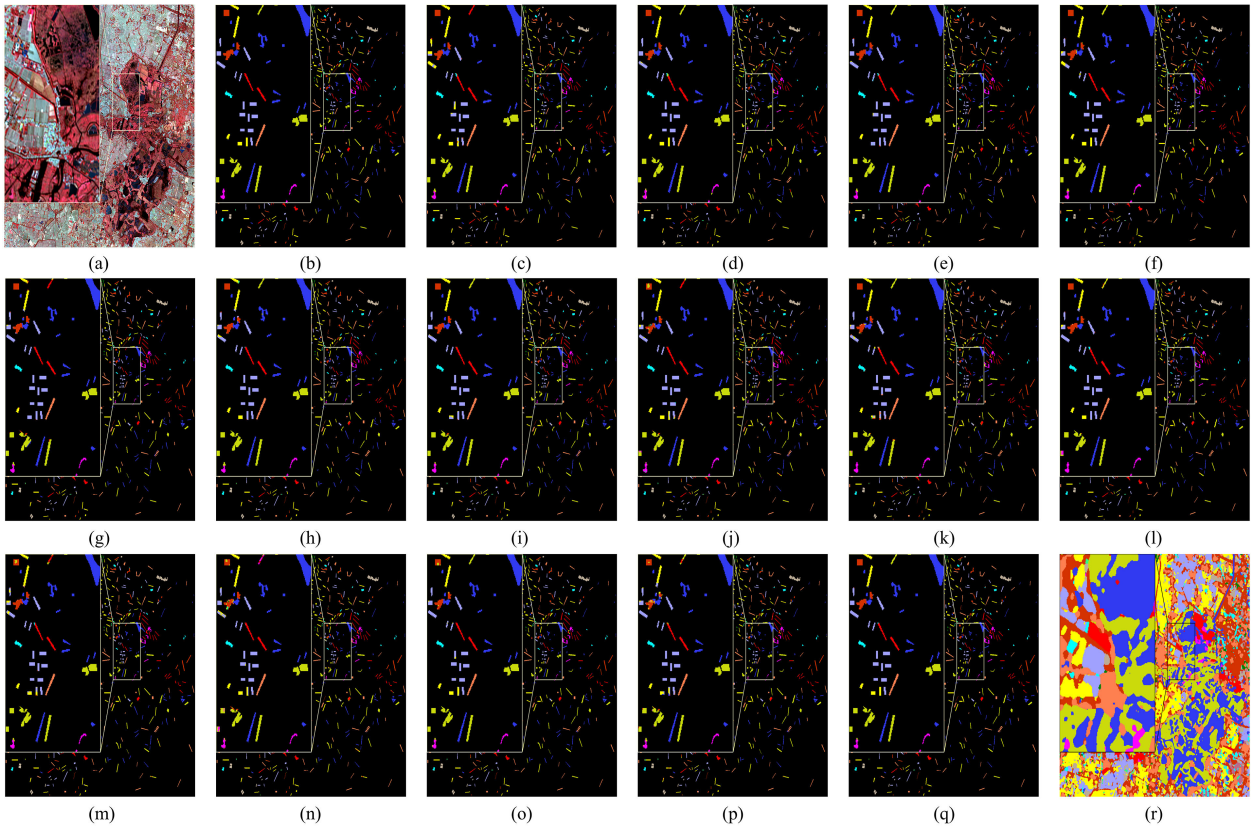


Fig. 6. Classification maps of the AW dataset. (a) False-color image. (b) Ground-truth map. (c) DHCN (OA = 97.11%). (d) SSRN (OA = 97.94%). (e) DCRN (OA = 98.24%). (f) PDCN (OA = 97.59%). (g) DBMA (OA = 97.30%). (h) DBDA (OA = 97.74%). (i) A^2S^2K (OA = 98.05%). (j) RSSAN (OA = 96.36%). (k) DOCN (OA = 96.74%). (l) SPRLT (OA = 98.03%). (m) DMuCA (OA = 97.19%). (n) SpeFm (OA = 95.57%). (o) SSFTT (OA = 97.88%). (p) CvT (OA = 95.80%). (q) Proposed GTCT (OA = 98.65%). (r) All pixel classification map by our proposed GTCT.

2) Visual Evaluation: In addition to quantitative comparisons, Figs. 4–7 correspondingly display classification maps generated by different methods on the four datasets. Based on the visual comparison results, our proposed GTCT method provides satisfactory classification maps with more details but less noise. However, other networks (e.g., DHCN, RSSAN, and DMuCA) generate more noisy pixels within classification maps, especially on the challenging IP dataset. In addition, Fig. 8 shows the t-SNE visualization results of hidden features through various deep learning algorithms on four different datasets. Compared with other competing methods, it can be observed that our proposed GTCT has a stronger aggregated distribution with less cross-aggregation and misjudgment among various categories. In particular, these scattered points of similar features generated by the proposed GTCT are more concentrated, while the central clusters of different types

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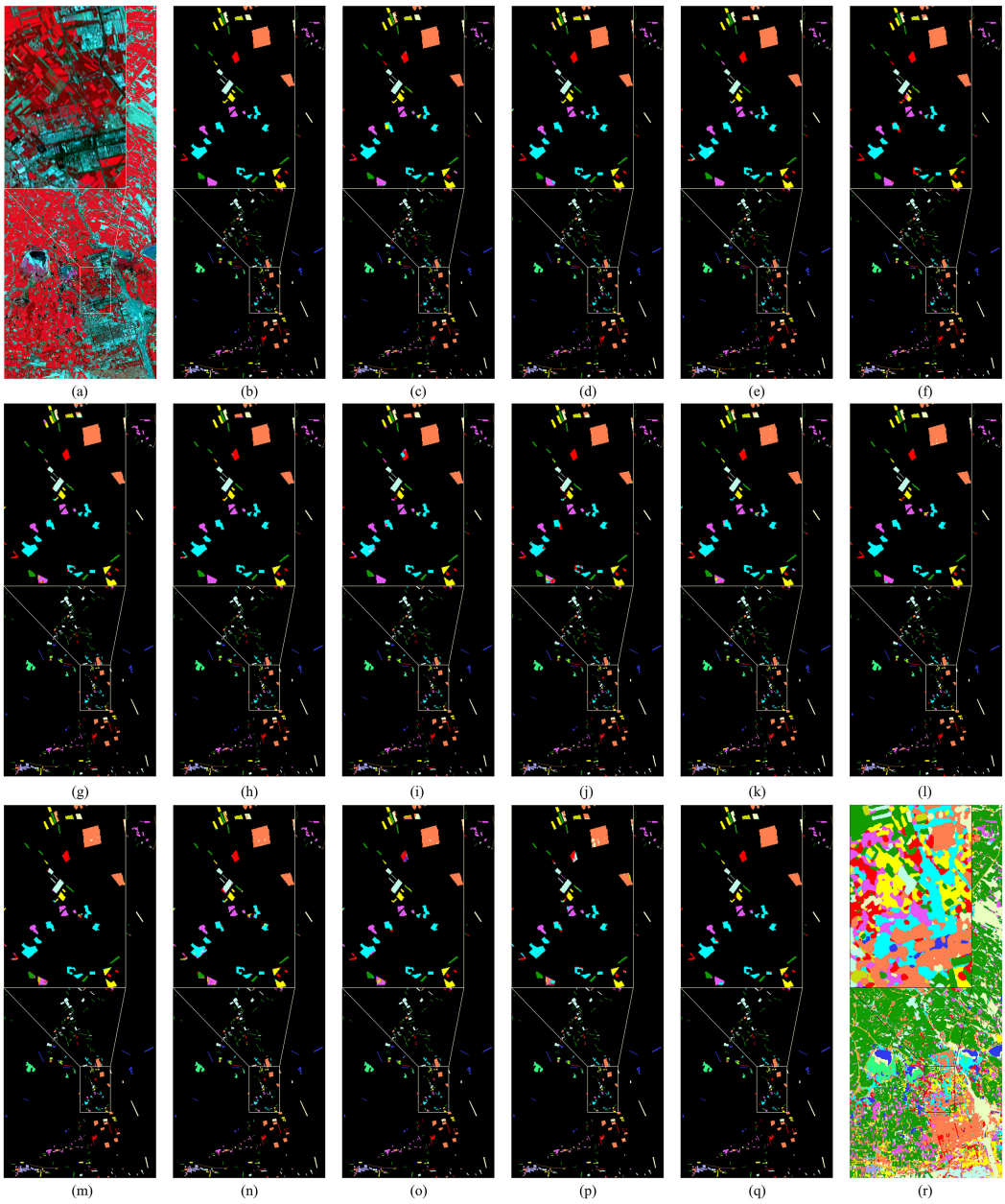


Fig. 7. Classification maps of the XC dataset. (a) False-color image. (b) Ground-truth map. (c) DHCN (OA = 94.86%). (d) SSRN (OA = 98.37%). (e) DCRN (OA = 98.38%). (f) PDCN (OA = 96.17%). (g) DBMA (OA = 96.72%). (h) DBDA (OA = 98.19%). (i) A^2S^2K (OA = 98.19%). (j) RSSAN (OA = 94.06%). (k) DOCN (OA = 95.11%). (l) SPRLT (OA = 97.10%). (m) DMuCA (OA = 94.96%). (n) SpeFm (OA = 91.38%). (o) SSFTT (OA = 96.62%). (p) CvT (OA = 91.56%). (q) Proposed GTCT (OA = 98.80%). (r) All pixel classification map by our proposed GTCT.

of features are farther apart. It also implies that the proposed GTCT can effectively model global dependencies and local relationships of different land covers from hyperspectral datasets.

Taking into account all comparison algorithms, the visual analyses are in consistent with the quantitative comparisons, validating the robustness and generalizability of GTCT under different classification scenarios. This is because our proposed GTCT is conductive to capture and integrate more heterogeneous and discriminative features from the spatial and spectral dimensions of HSI, where two types of global–local convolutional transformer encoders have a considerable effect on boosting classification effectiveness and reducing interfering pixels in classification maps.

3) Model Complexity: We objectively evaluate model complexity from both spatial and temporal aspects, where the number of parameters, average runtime, and FLOPs are compared and recorded in Fig. 9 and Fig. 10, respectively. In terms of space complexity, we can clearly observe that the proposed GTCT contains the least parameters among all models (e.g., 191 K on IP and 166 K on XC) on the four datasets. In general, CNN-based methods have more parameters than attention-based ones. The recently improved transformer-based models (e.g., SSFTT and SPRLT) typically contain fewer parameters than the vanilla transformer model (e.g., SpeFm). From the perspective of time complexity, the proposed GTCT also approaches the optimal SSFTT and surpasses other remaining comparison models in terms of FLOPs on all

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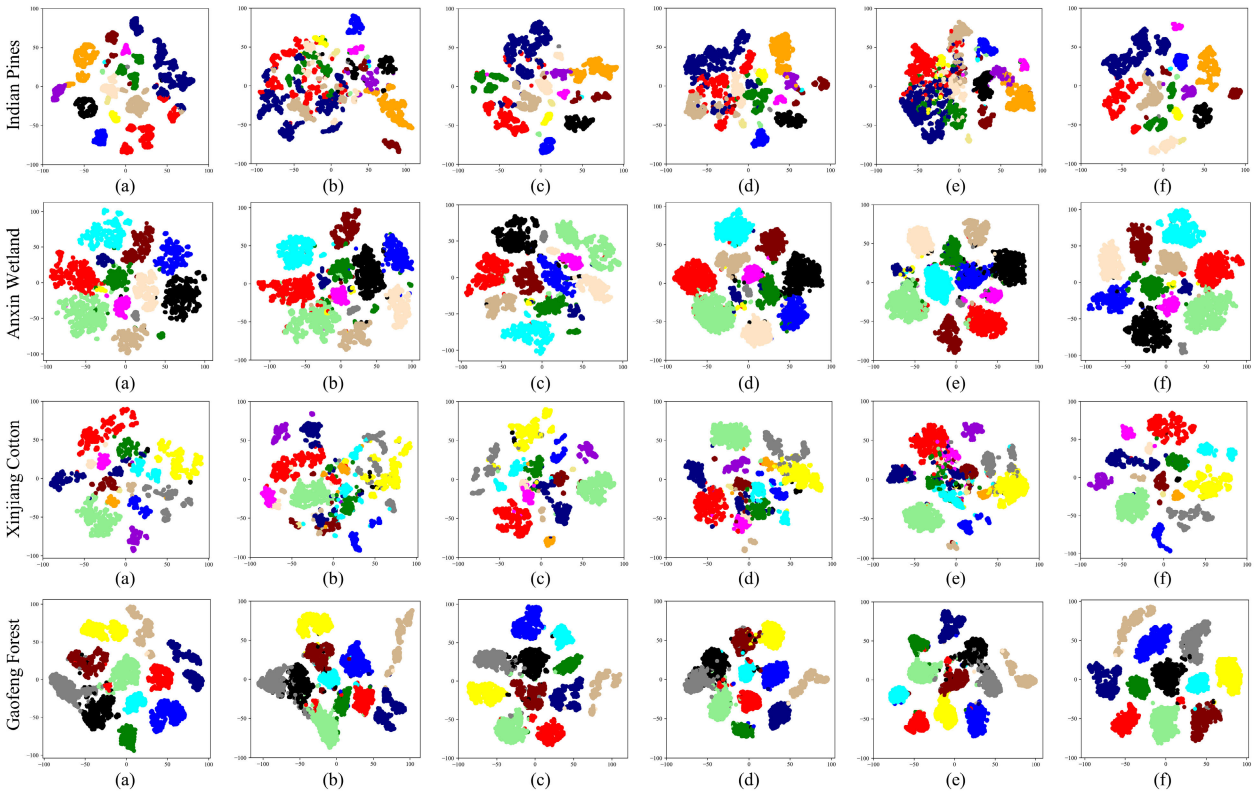


Fig. 8. t-SNE visualization results of hidden features produced by different methods on the four datasets. (a) DCRN. (b) DBMA. (c) SPRLT. (d) DMuCA. (e) SpeFm. (f) Proposed GTCT.

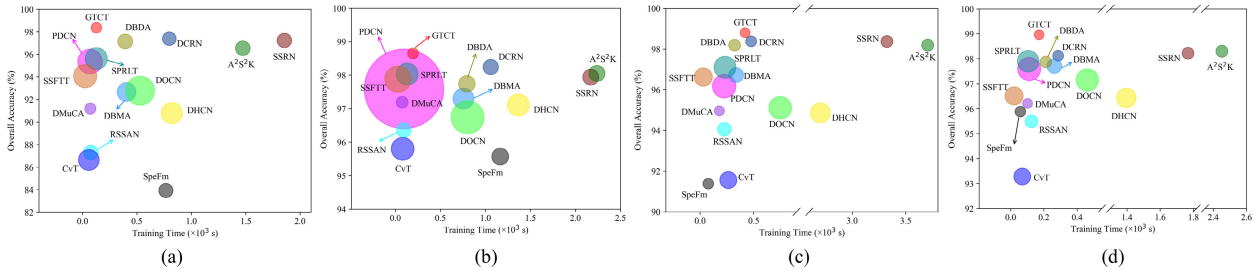


Fig. 9. Overall accuracy (%) on the four datasets compared to other methods with respect to model parameters. Larger marker size indicates more parameters. (a) IP. (b) AW. (c) XC. (d) GF.

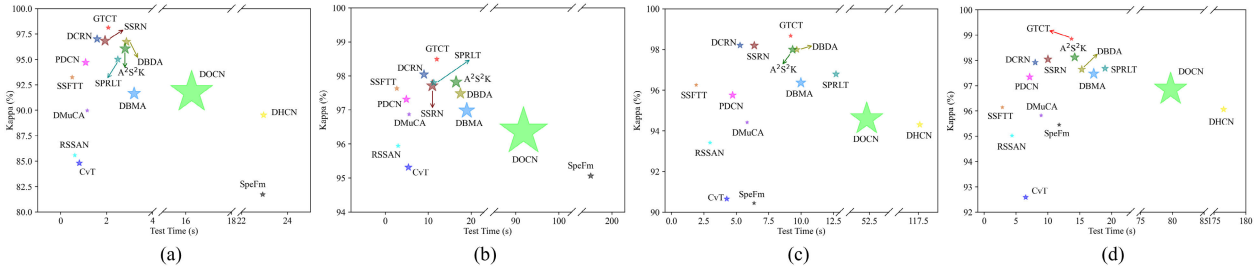


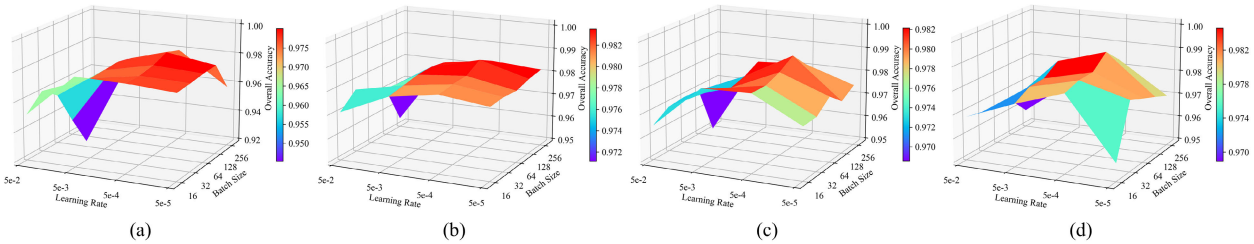
Fig. 10. Kappa (%) on the four datasets compared with other methods with respect to model FLOPs. Larger marker size indicates higher FLOPs. (a) IP. (b) AW. (c) XC. (d) GF.

datasets. This shows that the computational complexity of GTCT is also very low. For example, the GTCT has 2.5 G FLOPs with 2.0 G FLOPs for SSFTT on the IP dataset. For a model running time, the GTCT model outperforms state-of-the-art models (e.g., SSRN, DBDA, and A^{2}S^{2}K) in both training and testing stages for most HSI classification tasks. It is worth noting that, compared with CNN-based methods, the proposed method can greatly improve computational efficiency, where the training time of the SSRN model (1855.36 s) is ten times that of our GTCT model (128.37 s). The main reason is that CNN-based models usually take a relatively long time to train a large number of parameters. In summary,

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TABLE VII
ABLATION ANALYSIS OF THE PROPOSED METHOD ON THE FOUR DATASETS

Cases	Component					Overall Accuracy (%)			
	GECT	GACT	WMSFI	SSGAM	Polyloss	IP	GF	XC	AW
1	×	✓	✓	✓	✓	95.86±1.16	97.13±0.41	95.24±0.54	97.37±0.38
2	✓	×	✓	✓	✓	97.84±0.37	98.59±0.28	98.17±0.20	98.26±0.20
3	✓	✓	×	✓	✓	96.94±0.78	98.57±0.32	97.83±0.60	98.33±0.17
4	✓	✓	✓	×	✓	97.96±0.27	98.63±0.23	98.11±0.53	98.35±0.11
5	✓	✓	✓	✓	×	98.17±0.21	98.60±0.22	98.41±0.26	98.42±0.19
6	✓	✓	✓	✓	✓	**98.35±0.14**	**98.96±0.07**	**98.80±0.09**	**98.65±0.08**



(a) (b) (c) (d)

Fig. 11. Classification performance of GTCT with different learning rates and batch sizes on the four datasets. (a) IP. (b) AW. (c) XC. (d) GF.

with the least parameters and lower FLOPs, the proposed GTCT is capable of attaining state-of-the-art performance efficiently for vegetation fine mapping using large-scene satellite HSIs.

D. Ablation Studies

The essential ingredients to be ablated include the GECT module, GACT module, WMSFI module, SSGAM block, and Polyloss function. The experimental results of ablation studies are summarized in Table VII. When separately global–local spatial information is extracted by GACT, classification performance is significantly declined with OA dropping by 3.6% at most. It fully indicates that subtle spectral information is very important for discriminating vegetation subclasses with similar spectral characteristics. After removing the GACT module, inadequate utilization of spatial information will also weaken classification performance, especially for the fine classification task of various vegetation in a small area, e.g., on the IP dataset. In addition, the WMSFI enhances spectral–spatial feature interaction and integration adaptively across multiple scales, contributing significantly to the improvement of classification performance on the IP and XC datasets. The SSGAM block performs feature extraction from the spectral and spatial dimensions sequentially, further boosting the jointly spectral–spatial classification performance. We observe that simply replacing the widely used cross entropy loss with Polyloss can also slightly improve classification accuracy. These results validate the effectiveness of the components designed in the proposed GTCT.

E. Parameter Sensitivity Analysis

In this section, we investigate the optimal parameter configurations of the proposed GTCT method, where the proportions of training samples, learning rate, batch size, patch size, and the number of attention heads are analyzed.

1) Learning Rate and Batch Size: It is a fact that the learning rate directly affects the gradient descent rate. Meanwhile, an appropriate batch size can improve optimization convergence and reduce memory footprint. And the learning rate is usually scaled with the batch size accordingly for stable and reliable model training. Therefore, we carry out mixed experiments to analyze the mutual effect of learning rate and batch size on the classification performance of four hyperspectral datasets. For our proposed GTCT, we change the learning rate proportionally from 5e-5 to 5e-2 at ten times intervals, while the batch size is varied proportionally from 16 to 256 at two times intervals. As can be seen from Fig. 11, with a batch size of 128, the best classification performance can be obtained on the four datasets when the learning rate is set to 5e-4.

2) Patch Size: The spatial region of image patches has an immediate impact on the effective use of abundant spatial information. For four datasets, the spatial sizes of patches vary from 5 × 5 to 13 × 13 at intervals of 2. As shown in Fig. 12, the precision curves gradually rise first and then fall, achieving the highest classification accuracy when the patch size is almost 9 × 9. The findings indicate that smaller or larger image patches may lead to poor classification performance with a larger standard deviation. Finally, with a spatial size of 9 × 9, cubes are cropped as input for training all models to achieve optimal classification performance on four hyperspectral datasets.

3) Attention Heads: In the global–local feature encoding stages, the number of heads in spectral or spatial residual attention blocks (i.e., SpeRA and SpaRA) plays an important role in extracting representative and diverse features from HSI in different subspaces. Thus, to assess the impact of the different number of attention heads on the classification performance, the number of heads in two convolutional transformer encoders from spectral or spatial branches is set to {4, 6, 8, 12} and {2, 3, 4, 6}, respectively. It is worth noting that to visualize the performance differences of multiple algorithms more clearly, the overall accuracies under different parameter settings are processed by min–max normalization on the four datasets. As shown in Fig. 13, the experimental results show that the optimal parameters for attention heads on the four datasets are 8 and 4, respectively.

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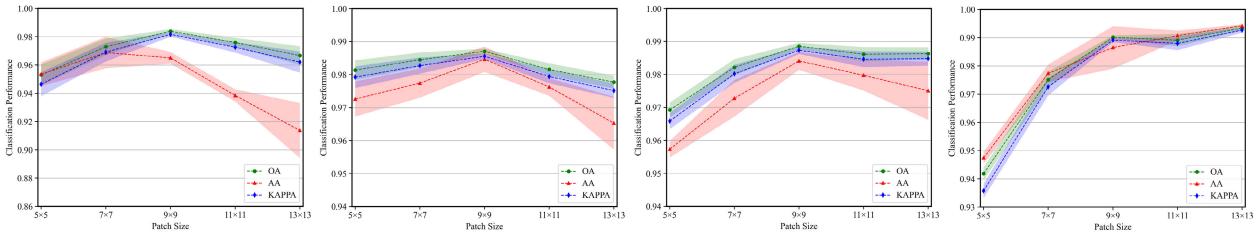


Fig. 12. Classification performance of GTCT with different spatial sizes of input patch on the four datasets. The shaded area refers to standard deviation of five experimental results. (a) IP. (b) AW. (c) XC. (d) GF.

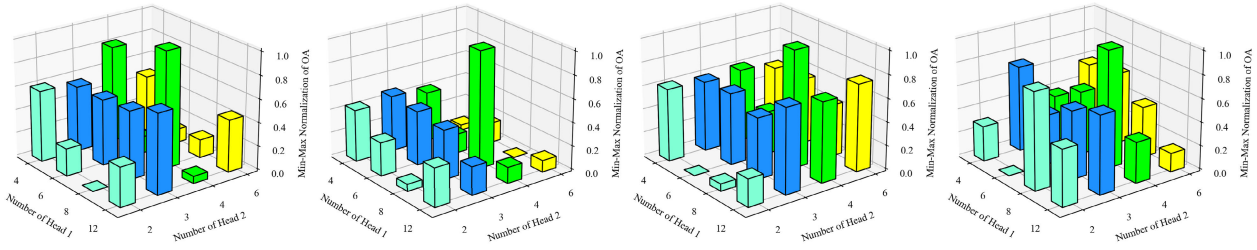


Fig. 13. Classification performance of GTCT with different numbers of attention heads on the four datasets. (a) IP. (b) AW. (c) XC. (d) GF.

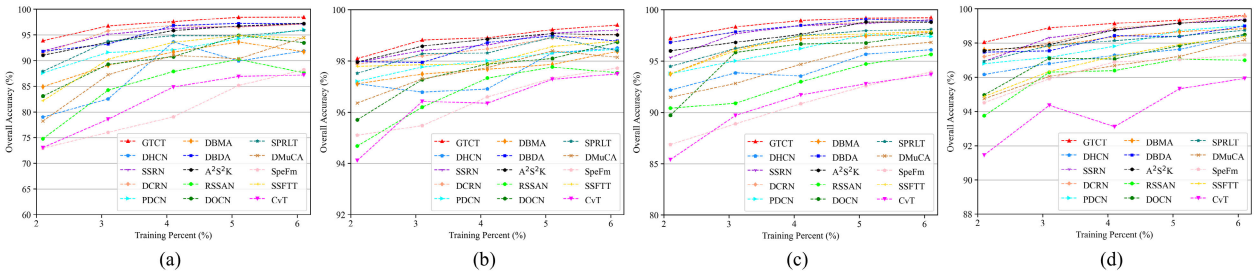


Fig. 14. Overall accuracy (%) of each method with different percentages of training samples on the four datasets. (a) IP. (b) AW. (c) XC. (d) GF.

4) Proportion of Training Samples: It is well known that increasing the number of training samples is helpful in improving the classification performance of deep learning models. We further conduct ablation experiments on different numbers of training samples by varying the proportions from 2% to 6% at intervals of 1%. And Fig. 14 reports the experimental results of different algorithms on the four datasets. As illustrated in Fig. 14, the OA increases gradually and steadily with more training samples on each dataset, achieving the best classification performance when a certain number of samples are trained. Compared with other methods, the classification performance of GTCT is almost always better than that of others under different sample ratios, indicating the stability and robustness of our proposed GTCT model. Taking runtime and classification performance into account, the reasonable proportions of training samples are 5% for IP, 3% for AW, 4% for XC, and 3% for GF, respectively.

V. Conclusion

In this work, we propose a novel GTCT for spectral–spatial classification of hyperspectral data, where two major advantages exist in this method. From a global and local perspective, 3-D convolution operation is introduced into the ViT architecture, either by replacing feature flattening with convolutional token embedding or integrating convolutional projection, rather than a linear projection into MHSA. As a result, the global dependencies among spectral bands or spatial pixels and local relationships that existed in adjacent bands or neighboring pixels are captured simultaneously. On the other hand, from the perspective of 3-D intrinsic characteristics of hyperspectral data, with different convolution kernels, the novel convolutional token embedding blocks and 3-D convolutional transformer encoders are investigated in both spectral and spatial dimensions to collaboratively capture discriminative features simultaneously. Accordingly, both global–local and spectral–spatial features are obtained. Thereafter, a weighted feature interaction module is exploited to interactively fuse and adaptively integrate the heterogeneous and multiscale features. Extensive experiments on the small-scale aerial and large-scene satellite hyperspectral datasets demonstrate that the proposed GTCT outperforms the other 14 state-of-the-art methods.

In future work, additional experiments with more complex classification scenarios should be conducted to further verify the generalizability and robustness of the proposed GTCT method. In addition, we will concentrate on automatically designing network architectures using neural architecture search techniques.

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REFERENCES

[1] X. Yang and Y. Yu, "Estimating soil salinity under various moisture conditions: An experimental study," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2525–2533, May 2017.

[2] L. Liang et al., "Estimation of crop LAI using hyperspectral vegetation indices and a hybrid inversion method," *Remote Sens. Environ.*, vol. 165, pp. 123–134, Aug. 2015.

[3] L. Zhang, L. Zhang, D. Tao, X. Huang, and B. Du, "Hyperspectral remote sensing image subpixel target detection based on supervised metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4955–4965, Aug. 2014.

[4] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[5] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[6] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4085–4098, Nov. 2010.

[7] L. Mou, P. Ghamisi, and X. X. Zhu, "Unsupervised spectral–spatial feature learning via deep residual Conv–Deconv network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 391–406, Jan. 2018.

[8] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[9] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[10] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jul. 2015.

[11] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[12] Y. Zhan, D. Hu, Y. Wang, and X. Yu, "Semisupervised hyperspectral image classification based on generative adversarial networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 212–216, Feb. 2018.

[13] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[14] C. Shi, J. Sun, T. Wang, and L. Wang, "Hyperspectral image classification based on a 3D octave convolution and 3D multiscale spatial attention network," *Remote Sens.*, vol. 15, no. 1, p. 257, Jan. 2023.

[15] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[16] K. Han, Y. Wang, H. Chen, X. Chen, and D. Tao, "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Dec. 2020.

[17] A. A. Aleissaee et al., "Transformers in remote sensing: A survey," 2022, *arXiv:2209.01206*.

[18] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[19] Z. Zhao, D. Hu, H. Wang, and X. Yu, "Convolutional transformer network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[20] R. Song, Y. Feng, W. Cheng, Z. Mu, and X. Wang, "BS2T: Bottleneck spatial–spectral transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532117.

[21] L. Yang et al., "FusionNet: A convolution–transformer fusion network for hyperspectral image classification," *Remote Sens.*, vol. 14, no. 16, p. 4066, Aug. 2022.

[22] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[23] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[24] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.

[25] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[26] Z. Meng, F. Zhao, M. Liang, and W. Xie, "Deep residual involution network for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 16, p. 3055, Aug. 2021.

[27] W. Qi, X. Zhang, N. Wang, M. Zhang, and Y. Cen, "A spectral–spatial cascaded 3D convolutional neural network with a convolutional long short-term memory network for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 20, p. 2363, Oct. 2019.

[28] H. Huang, C. Pu, Y. Li, and Y. Duan, "Adaptive residual convolutional neural network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2520–2531, 2020.

[29] Y. Xu et al., "Dual-channel residual network for hyperspectral image classification with noisy labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502511.

[30] J. Zhu, L. Fang, and P. Ghamisi, "Deformable convolutional neural networks for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 8, pp. 1254–1258, Aug. 2018.

[31] F. Zhao, J. Zhang, Z. Meng, and H. Liu, "Densely connected pyramidal dilated convolutional network for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 17, p. 3396, Aug. 2021.

[32] Z. Xue, X. Yu, B. Liu, X. Tan, and X. Wei, "HResNetAM: Hierarchical residual network with attention mechanism for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3566–3580, 2021.

[33] L. Zhao, J. Yi, X. Li, W. Hu, J. Wu, and G. Zhang, "Compact band weighting module based on attention-driven for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9540–9552, Nov. 2021.

[34] X. Zhang, S. Shang, X. Tang, J. Feng, and L. Jiao, "Spectral partitioning residual network with spatial attention mechanism for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5507714.

[35] W. Ma, Q. Yang, Y. Wu, W. Zhao, and X. Zhang, "Double-branch multi-attention mechanism network for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 11, p. 1307, Jun. 2019.

[36] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020.

[37] X. Tang et al., "Hyperspectral image classification based on 3-D octave convolution with spatial spectral attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2430–2447, Mar. 2021.

[38] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[39] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Jan. 2020.

[40] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[41] Z. Xue, Q. Xu, and M. Zhang, "Local transformer with spatial partition restore for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4307–4325, 2022.

[42] X. Tan, K. Gao, B. Liu, Y. Fu, and L. Kang, "Deep global-local transformer network combined with extended morphological profiles for hyperspectral image classification," *J. Appl. Remote Sens.*, vol. 15, no. 3, Sep. 2021, Art. no. 038509.

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[43] K. Liu et al., "Mapping coastal wetlands using transformer in transformer deep network on China ZY1–02D hyperspectral satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3891–3903, 2022.
[44] X. Hu, T. Li, T. Zhou, Y. Liu, and Y. Peng, "Contrastive learning based on transformer for hyperspectral image classification," *Appl. Sci.*, vol. 11, no. 18, p. 8670, Sep. 2021.
[45] Y. Gao et al., "Fusion classification of HSI and MSI using a spatial–spectral vision transformer for wetland biodiversity estimation," *Remote Sens.*, vol. 14, no. 4, p. 850, Feb. 2022.
[46] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528715.
[47] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514715.
[48] H. Wu et al., "CvT: Introducing convolutions to vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 22–31.
[49] M. Liang, Q. He, X. Yu, H. Wang, Z. Meng, and L. Jiao, "A dual multi-head contextual attention network for hyperspectral image classification," *Remote Sens.*, vol. 14, no. 13, p. 3091, Jun. 2022.
[50] J. Zhang, Z. Meng, F. Zhao, H. Liu, and Z. Chang, "Convolution transformer mixer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[51] Y. Liu, Z. Shao, and N. Hoffmann, "Global attention mechanism: Retain information to enhance channel-spatial interactions," Dec. 2021, *arXiv:2112.05561*. [Online]. Available: https://ui.adsabs.harvard.edu/abs/2021arXiv211205561L, doi: 10.48550/arXiv.2112.05561.
[52] R. O. Green et al., "Imaging spectroscopy and the airborne visible/infrared imaging spectrometer (AVIRIS)," *Remote Sens. Environ.*, vol. 65, no. 3, pp. 227–248, Sep. 1998.
[53] B. Zhang, L. Zhao, and X. Zhang, "Three-dimensional convolutional neural network model for tree species classification using airborne hyperspectral images," *Remote Sens. Environ.*, vol. 247, pp. 1–16, 2020.
[54] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," Nov. 2017, *arXiv:1711.05101*. [Online]. Available: https://ui.adsabs.harvard.edu/abs/2017arXiv171105101L, doi: 10.48550/arXiv.1711.05101.
[55] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.



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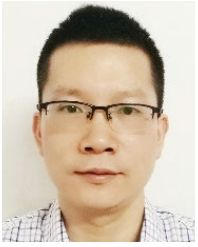
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